LOG-CONCAVITY AND LEVEL SET CURVATURE ESTIMATES FOR THE FIRST DIRICHLET EIGENFUNCTION IN HYPERBOLIC SPACE

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Abstract. In this paper, we study the first Dirichlet eigenfunction on bounded, smooth and strictly horo-convex domains in hyperbolic space $\mathbb{H}^n$. We prove its strict log-concavity in arbitrary dimension and establish the Gauss curvature bound $K > 1$ for its level sets. In dimensions two and three, we further obtain quantitative lower bounds for K in terms of the minimum Gauss curvature of the boundary. In higher dimensions, the product of the two smallest principal curvatures of each level hypersurface exceeds one. Furthermore, we extend the strict log-concavity result to bounded smooth horo-convex domains. Our approach combines maximum or minimum principles for suitable auxiliary functions, boundary convexity estimates and a domain deformation argument.

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1. Introduction

Given an elliptic boundary value problem on a bounded convex domain in a Riemannian manifold, does the solution inherit any convexity properties from the domain? This is a classical question in elliptic partial differential equations. There are two main aspects to such problems, namely the convexity of the solution (or of a suitable transformation of it) and the convexity of its level sets. A further goal is to obtain quantitative convexity estimates for solutions and in particular curvature estimates for their level sets in terms of the geometry of the boundary. The first Dirichlet eigenvalue problem and the torsion problem are two classical models for the study of these questions. For a bounded smooth domain Ω in a Riemannian manifold (M^n, g), these problems are given by

$$\begin{cases} \Delta u = -\lambda u & \text{in } \Omega, \\ u > 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{1.1}$$

and

$$\begin{cases} \Delta u = -2 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{1.2}$$

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respectively. Regarding the convexity of solutions to these two problems, we consider domains whose boundary principal curvatures are nonnegative. One studies the concavity of $\log u$ for the first Dirichlet eigenfunction u satisfying (1.1). For the torsion function u satisfying (1.2), the corresponding question concerns the convexity of $-\sqrt{u}$. In particular, both properties imply convexity of the superlevel sets of u.

In the Euclidean plane, Makar-Limanov [23] proved in 1971 that $-\sqrt{u}$ is strictly convex when u is the torsion function of a bounded convex domain. His proof is based on the superharmonicity of the P-function defined by

$$P = 2u \det \nabla^2 u + 2u_1 u_2 u_{12} - u_{11} u_2^2 - u_{22} u_1^2.$$

For the first Dirichlet eigenvalue problem, Brascamp and Lieb [5] established log-concavity of the first eigenfunction on bounded convex domains in $\mathbb{R}^n$. Acker, Payne and Philippin [1] subsequently adapted Makar-Limanov's idea to the two-dimensional eigenvalue problem to give another proof of log-concavity and obtain pointwise curvature estimates for the level curves. The study of log-concavity also has applications to fundamental gap estimates in spectral theory. In particular, Andrews and Clutterbuck [3] established a sharp modulus of concavity for the logarithm of the first eigenfunction and used it to prove the fundamental gap conjecture for convex Euclidean domains.

Several general methods have been developed to study the convexity properties of solutions to elliptic equations. Korevaar [17] introduced the concavity maximum principle for quasilinear elliptic and parabolic equations by applying the classical maximum principle to a suitable concavity function. Kennington [15] further refined this method and established power concavity results for a broad class of boundary value problems; see also the monograph of Kawohl [14]. An alternative approach developed by Alvarez, Lasry and Lions [2] uses convex envelopes together with the comparison principle and is sometimes referred to as the macroscopic convexity principle. The microscopic convexity principle, also known as the constant rank theorem, was introduced and developed through the work of Caffarelli and Friedman [6], Singer, Wong, Yau and Yau [26] and Korevaar and Lewis [19]. In particular, these approaches provide different ways to establish the convexity of $-\log u$ and $-\sqrt{u}$ for (1.1) and (1.2), respectively, on bounded convex domains in $\mathbb{R}^n$.

A natural question is whether the above convexity results extend to domains in a Riemannian manifold M^n. Convexity properties on space forms have also received considerable attention. In the following discussion, we focus on the first Dirichlet eigenvalue problem. For developments concerning the torsion problem, see our earlier work [29] and the references therein. When M^n is the sphere $\mathbb{S}^n$, Lee and Wang [20] proved log-concavity of the first Dirichlet eigenfunction on convex domains by adapting the continuity argument of Singer, Wong, Yau and Yau [26]. They also obtained a lower bound for the fundamental gap. In his survey article, Korevaar [18] reported unpublished joint work with Treibergs

extending the constant rank theorem to spaces of nonnegative constant curvature and indicated its application to the first eigenvalue problem on the sphere. Further progress on the sphere was made by Seto, Wang and Wei [24], who established improved log-concavity estimates and sharp fundamental gap bounds for $n \geq 3$ under a diameter restriction. He, Wei and Zhang [10] subsequently relaxed this restriction, while Dai, Seto and Wei [9] obtained the corresponding sharp fundamental gap bound for $n = 2$ using a different one-dimensional comparison model.

Beyond spaces of constant sectional curvature, Khan, Nguyen, Tuerkoen and Wei [16] obtained strong log-concavity estimates for the first eigenfunction on convex domains in positively curved surfaces under a condition involving the curvature and its derivatives, together with corresponding lower bounds for the fundamental gap. A classical example of a manifold with positive but nonconstant sectional curvature is complex projective space $\mathbb{CP}^n$, $n \geq 2$. In this setting, Xia [28] showed strict log-concavity of the first eigenfunction on smooth 1-convex domains in $\mathbb{CP}^2$ by establishing quantitative Hessian estimates with applications to level set convexity and the fundamental gap. Under a symmetry assumption, Hsu [11] proved strict log-concavity on smooth convex domains in $\mathbb{CP}^n$ for $n = 2, 3, 4$. More recently, Cho [7] established quantitative log-concavity and fundamental gap estimates in arbitrary dimension under a stronger boundary convexity assumption without symmetry restrictions.

The situation changes substantially in negative curvature. Shih [25] constructed a geodesically convex domain in the hyperbolic plane $\mathbb{H}^2$ whose first Dirichlet eigenfunction is not log-concave. Bourni, Clutterbuck, Nguyen, Stancu, Wei and Wheeler [4] subsequently proved that such counterexamples exist in $\mathbb{H}^n$ for every $n \geq 2$. These results show that geodesic convexity is insufficient to recover the corresponding Euclidean conclusions. A natural stronger geometric condition is horo-convexity. A domain $\Omega \subset \mathbb{H}^n$ is called horo-convex if every boundary point admits a supporting horoball containing Ω. For a smooth domain, this is equivalent to requiring that all boundary principal curvatures are at least 1. If they are all strictly greater than 1, the domain is called strictly horo-convex. In this setting, Wei and Xiao [27] used a constant rank theorem to establish log-concavity of the first eigenfunction on horo-convex domains of sufficiently small diameter in $\mathbb{H}^n$. In dimension two, Dai, Ennis, Nguyen and Wei [8] removed the diameter restriction and further proved that every superlevel set is horo-convex or equivalently that every nontrivial level curve has geodesic curvature at least 1. Their proof combines Killing fields with a nodal set argument.

Motivated by these developments, we study log-concavity of the first Dirichlet eigenfunction on horo-convex domains in hyperbolic space without a diameter restriction and obtain lower bounds for the curvature of its level sets.

Before stating our results, we briefly review the P-function approach to convexity problems in Euclidean space. In 2012, Ma, Shi and Ye [21] extended the arguments of Makar-Limanov [23] and Acker, Payne and Philippin [1] to higher dimensions by constructing suitable auxiliary functions and establishing minimum principles for them. More precisely, for the first Dirichlet eigenvalue problem, they considered the auxiliary function

$$\varphi = e^{-(n+1)v} \det \nabla^2 v,$$

where $v = -\log u$. This function is related to the Gauss curvature of the graph of v. Under the assumption $\nabla^2 v > 0$, they proved that φ is superharmonic modulo gradient terms with locally bounded coefficients. They then applied the minimum principle to obtain a lower bound for φ in terms of the boundary Gauss curvature and the boundary gradient of u. They also used this estimate in a deformation argument to prove log-concavity. Jia, Ma and Shi [13] further showed that the logarithm of this auxiliary function is superharmonic.

Inspired by this approach, we establish maximum or minimum principles for suitable auxiliary functions in hyperbolic space. We combine these principles with boundary convexity estimates and a domain deformation argument to prove the following results.

From now on, let $\Omega \subset \mathbb{H}^n$ be a bounded smooth domain and let u satisfy (1.1). Write $v = -\log u$. For $0 < t < \max_\Omega u$, denote the level sets of u by

$$\Sigma_t = \{x \in \Omega : u(x) = t\}.$$

At regular points, the principal curvatures of Σ_t are taken with respect to the outward normal of the superlevel set $\{u > t\}$. We use K to denote their product, namely the Gauss curvature. In dimension two, K is the geodesic curvature of the level curve.

Our first result concerns the two-dimensional case recently studied by Dai, Ennis, Nguyen and Wei [8]. Here we provide a new proof using PDE methods and derive a quantitative lower bound for the Gauss curvature of the level sets in terms of the minimum boundary curvature.

Theorem 1.1. *Assume that $n = 2$ and Ω is strictly horo-convex. Then, $\nabla^2 v > 0$ in Ω and the curvature of every level set Σ_t satisfies*

$$K \geq \min_{\partial\Omega} \kappa > 1.$$

In particular, every Σ_t is strictly horo-convex. Here κ denotes the curvature of $\partial\Omega$.

In fact, an approximation argument allows us to weaken the strict horo-convexity assumption on Ω to horo-convexity while retaining strict log-concavity of the first eigenfunction.

Corollary 1.2. *Assume that $n = 2$ and Ω is horo-convex. Then, $\nabla^2 v > 0$ in Ω.*

We now turn to higher dimensions, where strict log-concavity remains valid. In dimension three, the same Gauss curvature estimate holds as in the two-dimensional case with

the minimum Gauss curvature of $\partial\Omega$ as a lower bound. For $n \geq 4$, the product of the two smallest principal curvatures is strictly larger than 1, so the Gauss curvature also satisfies $K > 1$. More precisely, we prove the following result.

Theorem 1.3. *Let $n \geq 3$ and assume that Ω is strictly horo-convex. Then, $\nabla^2 v > 0$ in Ω. For $n = 3$, the Gauss curvature of every level set Σ_t satisfies*

$$K \geq \min_{\partial\Omega}(\kappa_1\kappa_2) > 1,$$

where κ_1, κ_2 are the principal curvatures of $\partial\Omega$. For $n \geq 4$, the two smallest principal curvatures $\kappa_1 \leq \kappa_2$ of Σ_t satisfy $\kappa_1\kappa_2 > 1$. In particular, $K > 1$ on every Σ_t.

The higher-dimensional counterpart of Corollary 1.2 is as follows.

Corollary 1.4. *Let $n \geq 3$ and suppose that Ω is horo-convex. Then, $\nabla^2 v > 0$ in Ω.*

Remark 1.5. *Similar results for the torsion problem (1.2) can also be established using similar methods. More precisely, $-\sqrt{u}$ can be shown to be strictly convex on bounded smooth strictly horo-convex domains in $\mathbb{H}^n$ without a diameter restriction. This removes the small diameter assumption in our earlier work [29].*

We now outline the proof. Our argument follows the auxiliary function approach developed by Ma, Shi and Ye [21]. The key is to construct suitable auxiliary functions and establish maximum or minimum principles for them. We then combine these principles with a domain deformation argument to prove strict log-concavity. For our choice of auxiliary functions ϕ, a boundary convexity estimate is also essential to keep $\nabla^2 v$ positive definite near the boundary throughout the deformation. The ball case establishes $\nabla^2 v > 0$ and $\phi < 1$. Smooth convergence of the eigenfunctions together with the boundary estimates ensures that these inequalities persist on domains sufficiently close to a ball. Proposition 2.5 provides the starting point for the continuity argument.

In dimension two, our choice of ϕ is motivated by the construction of Dai, Ennis, Nguyen and Wei [8]. Its boundary value is κ^{-2}, where κ denotes the curvature of $\partial\Omega$. We establish a maximum principle for ϕ and obtain $\phi \leq \max_{\partial\Omega} \kappa^{-2}$. Together with the strict eigenvalue bound $\lambda > \frac{1}{4}$, this estimate rules out the first degeneration of $\nabla^2 v$ along the deformation. On each regular level curve, the comparison $K^{-2} \leq \phi$ establishes the curvature estimate.

For the three-dimensional case, the auxiliary function has boundary value $(\kappa_1\kappa_2)^{-1}$, where κ_1, κ_2 are the principal curvatures of $\partial\Omega$. Geometrically, it is related to the area of a parallelogram spanned by an orthonormal tangent pair, with the area measured in the metric defined by the inverse Hessian; see Section 2.3 for details. A similar maximum principle controls ϕ by its boundary values. The strict bound $\lambda > 1$ allows us to exclude both possible ranks of a degenerate Hessian. The comparison $K^{-1} \leq \phi$ completes the Gauss curvature estimate.

For $n \geq 4$, we extend this construction by maximizing the same area quantity over all two-dimensional tangent planes of Σ_t. To control ϕ along the deformation, we establish a minimum principle for $e^{-v}(1-\phi)$. This provides a uniform upper bound for ϕ strictly below 1, which rules out Hessian degeneration and proves strict log-concavity. The level set curvature estimate follows from the same bound.

The rest of this paper is organized as follows. In Section 2, we collect the necessary preliminaries, including basic geometric facts, auxiliary functions, boundary convexity estimates and domain deformation. Sections 3, 4 and 5 contain the proofs of our main results in dimensions two, three and $n \geq 4$, respectively.

2. Preliminaries

This section begins with the geometry of level sets and the hyperbolic identities used below. We then introduce the auxiliary functions and explain their geometric meaning. Finally, we prove the ball and boundary estimates and describe the domain deformation used in the continuity argument.

2.1. Geometry of level sets.

We recall the formulas for the second fundamental form and the curvature of level sets; see Ma and Zhang [22]. Let g denote the hyperbolic metric on $\mathbb{H}^n$ and let ∇ be its Levi-Civita connection. We write $\langle \cdot, \cdot \rangle$ and $|\cdot|$ for the inner product and norm induced by g, respectively. Unless otherwise stated, Latin indices range from 1 to n and Greek indices range from 1 to $n-1$.

For a smooth function v, we denote the components of its first and second covariant derivatives in a local orthonormal frame $\{e_1, e_2, \cdots, e_n\}$ by

$$v_i = \nabla_{e_i} v, \quad v_{ij} = \nabla^2 v(e_i, e_j).$$

By the definition of the Hessian,

$$v_{ij} = \nabla_{e_i}(\nabla_{e_j} v) - \langle \nabla_{e_i} e_j, \nabla v \rangle . \tag{2.1}$$

Let u be the first Dirichlet eigenfunction satisfying (1.1) and set $v = -\log u$. For $0 < t < \max_\Omega u$, write

$$\Sigma_t = \{x \in \Omega : u(x) = t\} = \{x \in \Omega : v(x) = -\log t\}.$$

If t is a regular value of u, meaning that $\nabla u(x) \neq 0$ for every $x \in \Sigma_t$, then Σ_t is a smooth hypersurface by the implicit function theorem.

Since ∇u points into the superlevel set $\{u > t\}$ along Σ_t, the outward unit normal to Σ_t is

$$e_n = -\frac{\nabla u}{|\nabla u|} = \frac{\nabla v}{|\nabla v|}.$$

Along Σ_t, we complete e_n to a local orthonormal frame by choosing $e_1, e_2, \cdots, e_{n-1}$ tangent to Σ_t. As v is constant on Σ_t, we have

$$v_\alpha = 0, \quad v_n = \langle \nabla v, e_n \rangle = |\nabla v| > 0. \tag{2.2}$$

The second fundamental form of Σ_t with respect to e_n is defined by

$$h_{\alpha\beta} = \langle \nabla_{e_\alpha} e_n, e_\beta \rangle = - \langle \nabla_{e_\alpha} e_\beta, e_n \rangle . \tag{2.3}$$

Combining (2.1)-(2.3), we obtain

$$v_{\alpha\beta} = \nabla_{e_\alpha}(\nabla_{e_\beta} v) - \langle \nabla_{e_\alpha} e_\beta, \nabla v \rangle = -|\nabla v| \langle \nabla_{e_\alpha} e_\beta, e_n \rangle = |\nabla v| h_{\alpha\beta}.$$

Consequently,

$$h_{\alpha\beta} = \frac{v_{\alpha\beta}}{|\nabla v|}. \tag{2.4}$$

Let $\kappa_1, \kappa_2, \cdots, \kappa_{n-1}$ denote the principal curvatures of Σ_t, namely the eigenvalues of $(h_{\alpha\beta})$. The level hypersurface is strictly convex if and only if $(h_{\alpha\beta})$ is positive definite. In particular, (2.4) shows that if $\nabla^2 v > 0$, then every regular level set of v is strictly convex. We use K to denote the product of the principal curvatures, that is, the Gauss curvature of Σ_t. It follows from (2.4) that

$$K = \prod_{\alpha=1}^{n-1} \kappa_\alpha = \det(h_{\alpha\beta}) = \frac{\det(v_{\alpha\beta})}{|\nabla v|^{n-1}}. \tag{2.5}$$

2.2. **Hyperbolic geometry.** We next record the covariant derivative identities in hyperbolic space $\mathbb{H}^n$ that will be used in the proofs below. In a local orthonormal frame, we use the curvature convention

$$R_{ijkl} = -(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}).$$

For a smooth function u, we use subscripts to denote covariant derivatives, as in Section 2.1. The Ricci commutation identity gives

$$u_{ijk} - u_{ikj} = \delta_{ij} u_k - \delta_{ik} u_j. \tag{2.6}$$

In particular, for $i \neq j$,

$$u_{jii} = u_{iij} - u_j. \tag{2.7}$$

For fourth-order covariant derivatives, we have the commutation formula

$$\begin{aligned} u_{ijkl} - u_{ijlk} &= \sum_{m=1}^{n} (R_{mikl} u_{mj} + R_{mjkl} u_{im}) \\ &= \delta_{ik} u_{jl} - \delta_{il} u_{jk} + \delta_{jk} u_{il} - \delta_{jl} u_{ik}. \end{aligned}$$

Combining this with (2.6), we obtain for each fixed k,

$$u_{ijkk} = u_{kkij} - 2u_{ij} + 2\delta_{ij} u_{kk} - \delta_{ik} u_{jk} + \delta_{jk} u_{ik}. \tag{2.8}$$

For later use, we also recall the radial formulas in geodesic polar coordinates centered at $o \in \mathbb{H}^n$. Writing r for the geodesic distance from o, the metric takes the form

$$g = \mathrm{d}r^2 + \sinh^2 r \mathrm{d}z^2,$$

where $\mathrm{d}z^2$ denotes the standard metric on $\mathbb{S}^{n-1}$. For a radial function $u = u(r)$, its Laplacian is

$$\Delta u = u'' + (n-1)\coth r u', \tag{2.9}$$

and its Hessian is given by

$$\nabla^2 u = u'' \mathrm{d}r^2 + u' \coth r (g - \mathrm{d}r^2), \tag{2.10}$$

where $\mathrm{d}r^2 = \mathrm{d}r \otimes \mathrm{d}r$. In an orthonormal frame with $e_n = \partial_r$ and e_α tangent to the geodesic sphere centered at o, formula (2.10) gives

$$u_{\alpha\beta} = u' \coth r \delta_{\alpha\beta}, \quad u_{\alpha n} = 0, \quad u_{nn} = u'', \tag{2.11}$$

where $1 \le \alpha, \beta \le n-1$. Thus the radial and tangential eigenvalues of $\nabla^2 u$ are u'' and $u' \coth r$, respectively, with the latter having multiplicity $n-1$. If u is smooth at o, then $u'(0) = 0$ and $\nabla^2 u(o) = u''(0) g$.

We conclude this subsection with a basic lower bound for the first Dirichlet eigenvalue, which will play a key role in the subsequent arguments.

Lemma 2.1. *Suppose that u satisfies (1.1) on a bounded, smooth domain $\Omega \subset \mathbb{H}^n$, $n \ge 2$. Then, its first eigenvalue satisfies*

$$\lambda > \frac{(n-1)^2}{4}.$$

Proof. Choose geodesic polar coordinates with center $o \notin \bar{\Omega}$, so that r is smooth near $\bar{\Omega}$ and $\nabla r = \partial_r$ has unit length. By (2.9), we have

$$\Delta r = (n-1)\coth r > n-1.$$

Integrating with respect to the volume measure induced by g, we use the boundary condition and the Cauchy-Schwarz inequality to obtain

$$\begin{aligned}(n-1)\int_\Omega u^2 &< \int_\Omega u^2 \Delta r = -2\int_\Omega u \langle \nabla u, \nabla r \rangle \\ &\le 2\left(\int_\Omega u^2\right)^{1/2}\left(\int_\Omega |\nabla u|^2\right)^{1/2}.\end{aligned}$$

On the other hand, multiplying (1.1) by u and integrating over Ω yields

$$\int_\Omega |\nabla u|^2 = \lambda \int_\Omega u^2.$$

Consequently,

$$(n-1)\int_\Omega u^2 < 2\sqrt{\lambda}\int_\Omega u^2.$$

Since $u \not\equiv 0$, the asserted estimate follows. □

2.3. **Auxiliary functions.** We introduce the auxiliary functions used in the proofs of our main results and explain their geometric meaning. Their boundary limits reflect the curvature of $\partial\Omega$. Throughout, we work in the region where $\nabla^2 v > 0$.

We first consider the case $n = 2$ and define the auxiliary function

$$\phi = \frac{|(\nabla^2 v)\nabla v|^2}{(\det \nabla^2 v)^2}. \tag{2.12}$$

In terms of components, this takes the form

$$\phi = \frac{(v_{11}v_1 + v_{12}v_2)^2 + (v_{12}v_1 + v_{22}v_2)^2}{(v_{11}v_{22} - v_{12}^2)^2},$$

which is smooth wherever $\nabla^2 v > 0$ and vanishes at critical points of v. Its relation to the level set curvature can be seen as follows. At a regular point, let K be the curvature of the level curve and choose e_1 tangent to the level curve and $e_2 = \nabla v/|\nabla v|$. In this frame, $v_1 = 0$ and $v_2 = |\nabla v|$, and hence (2.4) gives

$$\phi = \frac{v_2^2(v_{12}^2 + v_{22}^2)}{(v_{11}v_{22} - v_{12}^2)^2} \geq \frac{v_2^2}{v_{11}^2} = \frac{1}{K^2}. \tag{2.13}$$

Our choice of ϕ is also motivated by the work of Dai, Ennis, Nguyen and Wei [8] on the horo-convexity of level curves. Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ denote rotation by $\pi/2$ in a local orthonormal frame. We can equivalently write

$$\phi = |(\nabla^2 v)^{-1} J \nabla v|^2,$$

using the two-dimensional identity $(\nabla^2 v)^{-1} J = J\nabla^2 v/\det \nabla^2 v$ and the fact that J preserves lengths. In their argument, after proving $\nabla^2 v > 0$, they use the vector $(\nabla^2 v)^{-1} J \nabla v$ to construct suitable Killing fields. If its length exceeds 1, the construction produces a function satisfying the same eigenvalue equation that vanishes together with its gradient at an interior point. However, horo-convexity ensures that the function has exactly two boundary zeros, which by the planar structure of nodal sets rules out such an interior zero. This contradiction yields

$$|(\nabla^2 v)^{-1} J \nabla v| \leq 1,$$

and consequently $K \geq 1$, proving the horo-convexity of the level curves.

In our approach, we apply a maximum principle to show that ϕ attains its maximum on $\partial\Omega$. Combining this with (2.13) and the boundary identity $\phi = \kappa^{-2}$ from Proposition 2.4, with κ denoting the curvature of $\partial\Omega$, we conclude that the curvature of every regular level curve is bounded below by the minimum curvature of $\partial\Omega$, that is,

$$K \geq \min_{\partial\Omega} \kappa.$$

We next consider the case $n = 3$, where the regular level sets are surfaces, and define

$$\phi = \frac{(\nabla v)^T(\nabla^2 v)\nabla v}{\det \nabla^2 v} = \frac{\sum_{i,j=1}^3 v_i v_j v_{ij}}{\det(v_{ij})}. \tag{2.14}$$

Here, ∇v is regarded as a column vector and $(\nabla v)^T$ denotes its transpose. As in the two-dimensional case, choose an orthonormal frame with e_1, e_2 tangent to a regular level surface and $e_3 = \nabla v/|\nabla v|$. The positive definiteness of the Hessian together with (2.5) implies

$$\phi = \frac{|\nabla v|^2 v_{33}}{\det(v_{ij})} \geq \frac{|\nabla v|^2}{v_{11}v_{22} - v_{12}^2} = \frac{1}{K}, \tag{2.15}$$

where K is the Gauss curvature of the level surface. The same maximum principle argument combined with the boundary identity $\phi = (\kappa_1\kappa_2)^{-1}$ established in Section 2.4 yields

$$K \geq \min_{\partial\Omega}(\kappa_1\kappa_2),$$

where κ_1, κ_2 denote the principal curvatures of $\partial\Omega$.

The auxiliary function ϕ defined in (2.14) also has a geometric interpretation in terms of tangent areas measured by the inverse Hessian. At each point p, the positive definite inverse matrix $(v^{ij}) = (v_{ij})^{-1}$ defines an inner product on the tangent space to $\mathbb{H}^3$ at p, given by $\sum_{i,j=1}^3 v^{ij}X_iY_j$, where X_i, Y_j are the components of tangent vectors $X, Y \in T_p\mathbb{H}^3$ in an orthonormal frame. Let ξ, η be local tangent vector fields along Σ_t orthonormal with respect to g and orthogonal to ∇v. In terms of their components ξ_i, η_i, (2.14) can then be expressed as

$$\phi = |\nabla v|^2 \left[\left(\sum_{i,j=1}^3 v^{ij}\xi_i\xi_j \right)\left(\sum_{i,j=1}^3 v^{ij}\eta_i\eta_j \right) - \left(\sum_{i,j=1}^3 v^{ij}\xi_i\eta_j \right)^2 \right]. \tag{2.16}$$

The expression in brackets is the squared area of the parallelogram spanned by ξ, η measured with respect to the metric defined by the inverse Hessian.

The area interpretation of the three-dimensional auxiliary function in (2.16) motivates the extension of the construction to higher dimensions. For $n \geq 4$, we maximize the same area quantity over all two-dimensional planes tangent to Σ_t and define

$$\phi = |\nabla v|^2 \max_{\substack{|\xi|=|\eta|=1, \langle\xi,\eta\rangle=0 \\ \langle\xi,\nabla v\rangle=\langle\eta,\nabla v\rangle=0}} \left[\left(\sum_{i,j=1}^n v^{ij}\xi_i\xi_j \right)\left(\sum_{i,j=1}^n v^{ij}\eta_i\eta_j \right) - \left(\sum_{i,j=1}^n v^{ij}\xi_i\eta_j \right)^2 \right]. \tag{2.17}$$

As in the three-dimensional case, this construction provides a curvature estimate. Let $0 < \kappa_1 \leq \cdots \leq \kappa_{n-1}$ be the principal curvatures of Σ_t. Choosing ξ, η in the principal directions corresponding to κ_1, κ_2, respectively, and using (2.4), we obtain

$$\phi \geq \frac{1}{\kappa_1\kappa_2}.$$

Following the strategy used in dimensions two and three, we combine a minimum principle for $e^{-v}(1-\phi)$ with a domain deformation to establish $\phi < 1$ in Ω, using the boundary identity

$$\phi = \frac{1}{\kappa_1 \kappa_2} < 1 \quad \text{on } \partial\Omega,$$

established in Section 2.4, where $\kappa_1, \kappa_2 > 1$ are the two smallest principal curvatures of $\partial\Omega$. Together with the preceding comparison, this yields

$$\kappa_1 \kappa_2 > 1$$

on every regular level hypersurface.

Finally, we define $\phi = 0$ at critical points of v, so that ϕ is continuous wherever $\nabla^2 v > 0$. Since the maximizing tangent plane may vary from point to point, ϕ need not be smooth. For the local calculations below, choose a smooth local orthonormal frame near a regular point x_0 such that the tangential projections of the first two directions span a maximizing plane at x_0. We orthonormalize these projections to obtain smooth tangent vector fields ξ, η. Substituting into the expression in (2.17), we calculate

$$\tilde{\phi} = |\nabla v|^2 \frac{\det\left(\sum_{i,j} v^{ij}\left(\delta_{ia} - \frac{v_i v_a}{|\nabla v|^2}\right)\left(\delta_{jb} - \frac{v_j v_b}{|\nabla v|^2}\right)\right)_{a,b=1,2}}{\det\left(\delta_{ab} - \frac{v_a v_b}{|\nabla v|^2}\right)_{a,b=1,2}}.$$

Moreover,

$$\tilde{\phi} = \frac{|\nabla v|^4}{\sum_{i=3}^n v_i^2} \det\left(v^{ab} - \frac{v_b}{|\nabla v|^2}\sum_i v^{ai} v_i - \frac{v_a}{|\nabla v|^2}\sum_i v^{bi} v_i + \frac{v_a v_b}{|\nabla v|^4}\sum_{i,j} v^{ij} v_i v_j\right)_{a,b=1,2}. \tag{2.18}$$

By construction, $\tilde{\phi}$ is smooth near x_0, agrees with ϕ at x_0 and is bounded above by ϕ nearby.

2.4. **The ball case and boundary convexity estimate.** To prepare for the proofs of the main theorems, we establish two convexity results for the first Dirichlet eigenfunctions, one on geodesic balls and the other near the boundary of horo-convex domains. The proofs are similar to those in [27] and [29]. We also compute the auxiliary functions introduced in Section 2.3 on geodesic balls and study their boundary behavior on horo-convex domains.

Let $B_R(o) \subset \mathbb{H}^n$ be the geodesic ball centered at o with radius R. We first prove Theorems 1.1 and 1.3 for geodesic balls, providing the results needed for the subsequent deformation argument.

Proposition 2.2. *Let u be the first Dirichlet eigenfunction satisfying (1.1) on $B_R(o)$. Then the function $v = -\log u$ satisfies $\nabla^2 v > 0$ in $B_R(o)$. Moreover, every level set Σ_t, $0 < t < \max_{B_R(o)} u$, is a geodesic sphere centered at o with all principal curvatures equal to $\coth r$, where r is its radius.*

Proof. By separation of variables, the first Dirichlet eigenfunction u on $B_R(o)$ is radial. Thus $v = -\log u$ depends only on r, the geodesic distance from o. By the radial Laplacian formula (2.9), the equation for v becomes

$$v'' + (n-1)\coth r v' = \lambda + (v')^2 \quad 0 < r < R. \tag{2.19}$$

Smoothness at the center gives $v'(0) = 0$. Taking the limit $r \to 0$ in (2.19) and using

$$\lim_{r\to 0} \coth r v'(r) = \lim_{r\to 0} (r\coth r)\frac{v'(r)}{r} = v''(0),$$

we obtain $v''(0) = \lambda/n > 0$.

Multiplying (2.19) by $\sinh^{n-1} r$ and integrating from 0 to r, we have

$$\sinh^{n-1} r v'(r) = \int_0^r \sinh^{n-1} s\big(\lambda + (v'(s))^2\big)\mathrm{d}s > 0.$$

Thus $v' > 0$ for $0 < r < R$. As noted in Section 2.2, the radial Hessian formula (2.10) shows that $\nabla^2 v$ has $n-1$ tangential eigenvalues equal to $v'\coth r$ and one radial eigenvalue v''. It remains to prove that $v'' > 0$.

Arguing by contradiction, suppose that v'' vanishes somewhere in $(0, R)$. Denoting its first zero by r_0, one has

$$v''(r) > 0 \quad \text{for } 0 \le r < r_0, \quad v''(r_0) = 0, \quad v'''(r_0) \le 0.$$

On the other hand, differentiating (2.19) gives

$$v''' = \big(2v' - (n-1)\coth r\big)v'' + \frac{n-1}{\sinh^2 r}v'. \tag{2.20}$$

Evaluating (2.20) at r_0, we obtain

$$v'''(r_0) = \frac{n-1}{\sinh^2 r_0}v'(r_0) > 0,$$

a contradiction. Hence $v'' > 0$ on $[0, R)$, and therefore $\nabla^2 v > 0$ in $B_R(o)$.

For the second assertion, radiality and strict monotonicity of v yield $\Sigma_t = \partial B_{r_t}(o)$ for some $r_t \in (0, R)$. By (2.10), in an orthonormal frame tangent to Σ_t, we have

$$h_{\alpha\beta} = \frac{v_{\alpha\beta}}{|\nabla v|} = \frac{v'(r_t)\coth r_t \delta_{\alpha\beta}}{v'(r_t)} = \coth r_t \delta_{\alpha\beta}.$$

Therefore, all its principal curvatures equal $\coth r_t$, completing the proof. □

On $B_R(o)$, we can explicitly evaluate the auxiliary functions introduced in Section 2.3. Applying the radial Hessian formula (2.11) to v, we obtain, in an orthonormal frame with $e_n = \partial_r$ and $e_1, e_2, \cdots, e_{n-1}$ tangent to the geodesic sphere centered at o,

$$v_n = v', \quad v_{\alpha\beta} = v'\coth r\delta_{\alpha\beta}, \quad v_{\alpha n} = 0, \quad v_{nn} = v''$$

for $0 < r < R$. Substituting these identities into (2.12) and (2.14) gives $\phi = \tanh^2 r$ when $n = 2, 3$. For $n \geq 4$, we have $v^{\alpha\beta} = (v' \coth r)^{-1}\delta_{\alpha\beta}$, and hence, for every orthonormal tangent pair ξ, η,

$$\sum_{i,j=1}^{n} v^{ij}\xi_i\xi_j = \sum_{i,j=1}^{n} v^{ij}\eta_i\eta_j = \frac{1}{v' \coth r}, \quad \sum_{i,j=1}^{n} v^{ij}\xi_i\eta_j = 0.$$

The definition of ϕ in (2.17) consequently yields $\phi = \tanh^2 r$. With $\phi(o) = 0$ at the center, we obtain in every dimension

$$0 \leq \phi < \tanh^2 R < 1 \quad \text{in } B_R(o). \tag{2.21}$$

We next show that $v = -\log u$ is strictly convex near the boundary of horo-convex domains.

Proposition 2.3. *Let Ω be a bounded, smooth horo-convex domain in $\mathbb{H}^n$ for $n \geq 2$, and let u be the first Dirichlet eigenfunction satisfying (1.1). Then, for $v = -\log u$, there exists a neighborhood $\mathcal{N}$ of $\partial\Omega$ such that $\nabla^2 v > 0$ in $\mathcal{N} \cap \Omega$.*

Proof. Let ν denote the unit outward normal to $\partial\Omega$. By the maximum principle and Hopf's lemma, we have $u > 0$ in Ω and $\nabla_\nu u < 0$ on $\partial\Omega$. Since $u = 0$ on $\partial\Omega$, all its tangential derivatives vanish there, and hence

$$\nabla u = (\nabla_\nu u)\nu = -|\nabla u|\nu \quad \text{on } \partial\Omega.$$

Choose an orthonormal frame $\{e_1, e_2, \cdots, e_{n-1}, \nu\}$ along $\partial\Omega$ and denote its second fundamental form by $h_{\alpha\beta}$. On $\partial\Omega$, we have

$$u_{\alpha\beta} = e_\alpha(e_\beta u) - \langle \nabla_{e_\alpha} e_\beta, \nabla u\rangle = h_{\alpha\beta}\nabla_\nu u = -|\nabla u| h_{\alpha\beta}.$$

Thus, for every unit tangential vector e, horo-convexity implies

$$-\nabla^2 u(e, e) = |\nabla u| h(e, e) \geq |\nabla u| > 0. \tag{2.22}$$

Since $v = -\log u$, direct computation in Ω yields

$$\nabla v = -\frac{\nabla u}{u}, \quad \nabla^2 v = \frac{\nabla u \otimes \nabla u}{u^2} - \frac{\nabla^2 u}{u}.$$

Therefore, for any unit vector e,

$$\nabla^2 v(e, e) = \frac{|\nabla_e u|^2}{u^2} - \frac{\nabla^2 u(e, e)}{u}. \tag{2.23}$$

Near $\partial\Omega$, write $e = e^\top + e^\perp$, where $e^\perp = \langle e, \nabla u/|\nabla u|\rangle \nabla u/|\nabla u|$. Then

$$|\nabla_e u|^2 = \langle e, \nabla u\rangle^2 = \left\langle e, |\nabla u|\frac{\nabla u}{|\nabla u|}\right\rangle^2 = |\nabla u|^2 |e^\perp|^2,$$

and substituting into (2.23), we obtain

$$\nabla^2 v(e, e) = \frac{|\nabla u|^2 |e^\perp|^2}{u^2} - \frac{\nabla^2 u(e, e)}{u}.$$

By (2.22), compactness and continuity, there exists $\eta > 0$ such that $\nabla^2 v(e,e) > 0$ near $\partial\Omega$ for every unit vector e satisfying $|e^\perp| \le \eta$. Since $u \to 0$ while $|\nabla u|$ is bounded away from zero by Hopf's lemma and $\nabla^2 u$ is bounded near $\partial\Omega$, we have $\nabla^2 v(e,e) \to +\infty$ uniformly for $|e^\perp| \ge \eta$ as $x \to \partial\Omega$. Hence $\nabla^2 v > 0$ in $\mathcal{N} \cap \Omega$ for a sufficiently small neighborhood $\mathcal{N}$ of $\partial\Omega$, which completes the proof. □

The preceding boundary convexity estimate guarantees that the auxiliary functions introduced in Section 2.3 are well defined near $\partial\Omega$. We now determine their boundary values.

Proposition 2.4. *Let $\Omega \subset \mathbb{H}^n$, $n \ge 2$, be a bounded, smooth and horo-convex domain, and let u be its first Dirichlet eigenfunction satisfying (1.1). Set $v = -\log u$ and let ϕ be the corresponding auxiliary function defined in (2.12), (2.14) or (2.17). Then ϕ extends continuously to $\partial\Omega$. In dimension two,*

$$\lim_{x\to p} \phi(x) = \frac{1}{\kappa(p)^2}, \quad p \in \partial\Omega, \tag{2.24}$$

where κ is the boundary curvature. For $n \ge 3$,

$$\lim_{x\to p} \phi(x) = \frac{1}{\kappa_1(p)\kappa_2(p)}, \quad p \in \partial\Omega, \tag{2.25}$$

where $\kappa_1(p) \le \kappa_2(p)$ are the two smallest principal curvatures of $\partial\Omega$ at p.

Proof. Fix $p \in \partial\Omega$. By Hopf's lemma and continuity, $\nabla u \ne 0$ near p. Choose a local orthonormal frame with $e_n = -\nabla u/|\nabla u|$ and $e_1, e_2, \cdots, e_{n-1}$ tangent to the level hypersurface of u, so that $u_\alpha = 0$ and $u_n = -|\nabla u|$. At p, choose the tangential vectors in the principal directions of $\partial\Omega$, ordered so that

$$1 \le \kappa_1(p) \le \kappa_2(p) \le \cdots \le \kappa_{n-1}(p).$$

On $\partial\Omega$, e_n is the outward unit normal, and (2.22) implies

$$-u_{\alpha\beta}(p) = |\nabla u(p)| h_{\alpha\beta}(p) = |\nabla u(p)| \kappa_\alpha(p) \delta_{\alpha\beta}. \tag{2.26}$$

Since $v = -\log u$ and $u_\alpha = 0$ in the chosen frame,

$$v_{\alpha\beta} = -\frac{u_{\alpha\beta}}{u}, \quad v_{\alpha n} = -\frac{u_{\alpha n}}{u}, \quad v_{nn} = \frac{u_n^2 - u u_{nn}}{u^2}.$$

In dimension two, $v_1 = 0$ and $v_2 = -u_2/u$, so (2.12) becomes

$$\phi = \frac{v_2^2(v_{12}^2 + v_{22}^2)}{(v_{11}v_{22} - v_{12}^2)^2} = \frac{u_2^2\big(u^2 u_{12}^2 + (u_2^2 - u u_{22})^2\big)}{\big(u_{11}(u_2^2 - u u_{22}) + u u_{12}^2\big)^2}.$$

As $x \to p$, we have $u \to 0$, $u_2(p) < 0$ and $u_{11}(p) = \kappa(p) u_2(p)$. Therefore,

$$\lim_{x\to p} \phi(x) = \frac{u_2(p)^2}{u_{11}(p)^2} = \frac{1}{\kappa(p)^2}.$$

We now consider $n \geq 3$. A direct calculation from $\sum_{k=1}^{n} v_{ik} v^{kj} = \delta_{ij}$ and $|\nabla v| = |\nabla u|/u$ shows that

$$(|\nabla v| v^{\alpha\beta})_{\alpha,\beta=1}^{n-1} = |\nabla u| \left(\left(-u_{\alpha\beta} - \frac{u u_{\alpha n} u_{\beta n}}{u_n^2 - u u_{nn}} \right)_{\alpha,\beta=1}^{n-1} \right)^{-1}.$$

Taking $x \to p$ and recalling that $u(p) = 0$ and $u_n(p) \neq 0$, we conclude from (2.26) that

$$\lim_{x \to p} (|\nabla v| v^{\alpha\beta})_{\alpha,\beta=1}^{n-1} = \operatorname{diag}\left(\frac{1}{\kappa_1(p)}, \frac{1}{\kappa_2(p)}, \cdots, \frac{1}{\kappa_{n-1}(p)} \right).$$

In particular, each component satisfies

$$\lim_{x \to p} (|\nabla v| v^{\alpha\beta}) = \frac{\delta_{\alpha\beta}}{\kappa_\alpha(p)}. \tag{2.27}$$

In dimension three, choosing $\xi = e_1$ and $\eta = e_2$ reduces (2.16) to

$$\phi = (|\nabla v| v^{11})(|\nabla v| v^{22}) - (|\nabla v| v^{12})^2.$$

It follows from (2.27) that

$$\lim_{x \to p} \phi(x) = \frac{1}{\kappa_1(p)\kappa_2(p)}.$$

For $n \geq 4$, the convergence in (2.27) is uniform for all orthogonal unit tangent vectors ξ and η. Hence, by (2.17),

$$\lim_{x \to p} \phi(x) = \max_{1 \leq \alpha < \beta \leq n-1} \frac{1}{\kappa_\alpha(p)\kappa_\beta(p)} = \frac{1}{\kappa_1(p)\kappa_2(p)},$$

where the maximum is attained in the first two principal directions. Since the boundary values depend continuously on p, ϕ extends continuously to $\partial\Omega$. This completes the proof. □

2.5. **Domain deformation.** To establish convexity, we use a continuity argument that relies on a deformation of the domain. We therefore need a smooth family of strictly horo-convex domains connecting the given domain to a geodesic ball. Following [27], we describe this deformation and establish the inequalities needed to begin the continuity argument.

Let $\Omega \subset \mathbb{H}^n$, $n \geq 2$, be a bounded, smooth and strictly horo-convex domain. For $n \geq 3$, strict horo-convexity ensures that $\partial\Omega$ is an admissible initial hypersurface for the locally constrained curvature flow of Hu, Li and Wei [12]. The flow produces a smooth family of strictly horo-convex domains $\{\Omega_t\}_{t \geq 0}$ with $\Omega_0 = \Omega$, whose boundaries converge smoothly to a geodesic sphere of radius $R > 0$ centered at $o \in \mathbb{H}^n$,

$$\partial\Omega_t \to \partial B_R(o) \quad \text{as } t \to \infty.$$

For $n = 2$, the corresponding family follows from the dimension reduction argument in [27].

Along this deformation, let u_t be the first Dirichlet eigenfunction satisfying (1.1) on Ω_t, normalized by $\int_{\Omega_t} u_t^2 = 1$, where integration is with respect to the volume measure induced

by g. Set $v_t = -\log u_t$ and let ϕ_t be the corresponding auxiliary function from Section 2.3, defined wherever $\nabla^2 v_t > 0$. By standard elliptic theory, the normalized eigenfunctions u_t depend smoothly on t and converge to the normalized first Dirichlet eigenfunction u_∞ on $B_R(o)$ with smooth convergence up to the boundary. In particular, v_t converges locally smoothly to $v_\infty = -\log u_\infty$ in the interior.

The convergence above together with the ball result and the boundary estimates allows us to begin the continuity argument through the following proposition.

Proposition 2.5. *There exists $T > 0$ such that $\nabla^2 v_t > 0$ and $\phi_t < 1$ in Ω_t for every $t \geq T$.*

Proof. Proposition 2.2 and (2.21) establish $\nabla^2 v_\infty > 0$ and $\phi_\infty < 1$ on $B_R(o)$, where ϕ_∞ is the auxiliary function corresponding to v_∞. Since v_t converges locally smoothly to v_∞, the same conclusions hold on each compact subset of $B_R(o)$ for all sufficiently large t.

Meanwhile, smooth convergence of the domains and eigenfunctions provides uniform derivative bounds for u_t and a positive lower bound for $|\nabla u_t|$ on $\partial\Omega_t$ for all sufficiently large t. These bounds allow us to apply the boundary convexity argument for $v_t = -\log u_t$ in Proposition 2.3 uniformly, so that $\nabla^2 v_t > 0$ in a boundary neighborhood of uniform width. Moreover, (2.24) and (2.25) together with the convergence of the boundary principal curvatures to $\coth R > 1$ imply that the boundary values of ϕ_t converge uniformly to $\tanh^2 R < 1$. The uniform bounds above and the formulas in the proof of Proposition 2.4 ensure that ϕ_t approaches its boundary values uniformly in t. Hence, $\phi_t < 1$ in a boundary neighborhood of uniform width.

Finally, the interior and boundary conclusions together show that there exists $T > 0$ such that $\nabla^2 v_t > 0$ and $\phi_t < 1$ throughout Ω_t for every $t \geq T$, completing the proof. □

3. The two-dimensional case

We begin with the two-dimensional case and prove Theorem 1.1, illustrating the general strategy for our main results. Let u be the first Dirichlet eigenfunction satisfying (1.1) on a bounded, smooth and strictly horo-convex domain $\Omega \subset \mathbb{H}^2$. The function $v = -\log u$ solves

$$\Delta v = \lambda + |\nabla v|^2 \quad \text{in } \Omega. \tag{3.1}$$

We use the auxiliary function ϕ defined in (2.12) and write κ for the boundary curvature. The proof proceeds by first establishing a maximum principle for ϕ under the assumptions $\nabla^2 v > 0$ and $\phi < 1$. We then apply the domain deformation from Section 2.5 and rule out the first degeneration of the Hessian, thereby establishing strict log-concavity. Moreover, the level set curvature estimate follows from this maximum principle.

3.1. **A maximum principle.** The following proposition bounds ϕ in terms of the boundary curvature.

Proposition 3.1. *Suppose that* $\nabla^2 v > 0$ *and* $\phi < 1$ *in* Ω. *Then*

$$\phi \le \max_{\partial\Omega} \kappa^{-2} < 1. \tag{3.2}$$

Proof. In view of (2.24), ϕ extends continuously to $\bar{\Omega}$ with positive boundary values κ^{-2}. To obtain the desired estimate (3.2), it suffices to show that ϕ attains its maximum on $\partial\Omega$. Suppose to the contrary that (3.2) fails. Then, ϕ attains its maximum at some $x_0 \in \Omega$, where $0 < \phi(x_0) < 1$, $\nabla\phi(x_0) = 0$ and $\Delta\phi(x_0) \le 0$.

Note that $\nabla|\nabla v|^2 = 2(\nabla^2 v)\nabla v$, so $\phi(x_0) > 0$ implies $\nabla|\nabla v|^2(x_0) \neq 0$. Choose orthonormal coordinates at x_0 whose second coordinate vector points in the direction of $\nabla|\nabla v|^2$. All the following calculations are carried out at x_0, and all indices range from 1 to 2. By the choice of coordinates,

$$v_1 v_{11} + v_2 v_{12} = 0, \tag{3.3}$$

so the positive definiteness of $\nabla^2 v$ yields

$$v_1 = -\frac{v_{12}}{v_{11}} v_2, \quad v_2 > 0. \tag{3.4}$$

Consequently,

$$v_1 v_{12} + v_2 v_{22} = \frac{v_2}{v_{11}}(v_{11}v_{22} - v_{12}^2). \tag{3.5}$$

From (3.3) and (3.5), the expression for ϕ reduces to

$$\phi = \frac{v_2^2}{v_{11}^2}. \tag{3.6}$$

Differentiating the definition of ϕ in (2.12) and using $\nabla\phi(x_0) = 0$, we find

$$\begin{aligned} 0 = \frac{\phi_k}{2\phi} = \frac{v_1 v_{12k} + v_2 v_{22k} + v_{12} v_{1k} + v_{22} v_{2k}}{v_1 v_{12} + v_2 v_{22}} \\ - \frac{v_{22} v_{11k} + v_{11} v_{22k} - 2 v_{12} v_{12k}}{v_{11} v_{22} - v_{12}^2}. \end{aligned} \tag{3.7}$$

The relations (3.4) and (3.5) simplify (3.7) to

$$v_{22} v_{11k} - v_{12} v_{12k} = \frac{v_{11}}{v_2}(v_{12} v_{1k} + v_{22} v_{2k}). \tag{3.8}$$

Taking derivatives once more, summing over k and using $\phi_k = 0$, we have

$$\begin{aligned}\frac{\Delta\phi}{2\phi} =& -\frac{1}{v_{11}v_{22}-v_{12}^2}\sum_k (v_{22}v_{11kk}+v_{11}v_{22kk}-2v_{12}v_{12kk}+2v_{11k}v_{22k}-2v_{12k}^2)\\ &+\frac{1}{(v_1v_{12}+v_2v_{22})^2}\sum_k (v_1v_{11k}+v_2v_{12k}+v_{11}v_{1k}+v_{12}v_{2k})^2\\ &+\frac{1}{v_1v_{12}+v_2v_{22}}\sum_k (v_1v_{12kk}+v_2v_{22kk}+2v_{12k}v_{1k}+2v_{22k}v_{2k}\\ &\qquad +v_{12}v_{1kk}+v_{22}v_{2kk}).\end{aligned} \tag{3.9}$$

To handle the fourth-order derivative terms in (3.9), we differentiate the equation (3.1) and use (2.7) to write

$$\sum_k v_{ikk} = (\Delta v)_i - v_i = 2\sum_k v_k v_{ki} - v_i. \tag{3.10}$$

The second derivatives of (3.1) together with (2.6) yield

$$\begin{aligned}(\Delta v)_{ij} &= 2\sum_k v_{ki}v_{kj} + 2\sum_k v_k v_{kij}\\ &= 2\sum_k v_{ki}v_{kj} + 2\sum_k v_k v_{ijk} + 2v_iv_j - 2\delta_{ij}\sum_k v_k^2.\end{aligned}$$

It follows from (2.6) and (2.8) that

$$\begin{aligned}\sum_k v_{ijkk} &= (\Delta v)_{ij} - 4v_{ij} + 2\delta_{ij}\Delta v\\ &= 2\sum_k v_k v_{ijk} + 2\sum_k v_{ki}v_{kj} + 2v_iv_j - 4v_{ij} + 2\lambda\delta_{ij}.\end{aligned} \tag{3.11}$$

Using the relations (3.4) and (3.5), we calculate the fourth-order derivative terms in (3.9) as

$$\begin{aligned}&-\frac{1}{v_{11}v_{22}-v_{12}^2}\sum_k (v_{22}v_{11kk}+v_{11}v_{22kk}-2v_{12}v_{12kk})\\ &+\frac{1}{v_1v_{12}+v_2v_{22}}\sum_k (v_1v_{12kk}+v_2v_{22kk})\\ =&-\frac{1}{v_{11}v_{22}-v_{12}^2}\sum_k (v_{22}v_{11kk}-v_{12}v_{12kk})\\ =&-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k v_k(v_{22}v_{11k}-v_{12}v_{12k})\\ &-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k (v_{22}v_{1k}^2-v_{12}v_{1k}v_{2k})\end{aligned}$$

$$-\frac{2(v_1^2v_{22}-v_1v_2v_{12})}{v_{11}v_{22}-v_{12}^2}+4-\frac{2\lambda v_{22}}{v_{11}v_{22}-v_{12}^2},$$

where (3.11) has been used in the last equality. Applying (3.5), we rewrite (3.9) as

$$\begin{aligned}\frac{\Delta\phi}{2\phi}=&-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k v_k(v_{22}v_{11k}-v_{12}v_{12k})\\&-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k(v_{11k}v_{22k}-v_{12k}^2)\\&+\frac{v_{11}^2}{v_2^2(v_{11}v_{22}-v_{12}^2)^2}\sum_k(v_1v_{11k}+v_2v_{12k}+v_{11}v_{1k}+v_{12}v_{2k})^2\\&+\frac{2v_{11}}{v_2(v_{11}v_{22}-v_{12}^2)}\sum_k(v_{1k}v_{12k}+v_{2k}v_{22k})\\&+\frac{v_{11}}{v_2(v_{11}v_{22}-v_{12}^2)}\sum_k(v_{12}v_{1kk}+v_{22}v_{2kk})\\&-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k(v_{22}v_{1k}^2-v_{12}v_{1k}v_{2k})\\&-\frac{2(v_1^2v_{22}-v_1v_2v_{12})}{v_{11}v_{22}-v_{12}^2}+4-\frac{2\lambda v_{22}}{v_{11}v_{22}-v_{12}^2}.\end{aligned}\tag{3.12}$$

We next deal with the third-order derivative terms in (3.12). The relations (3.8) and (3.10) imply

$$\begin{aligned}&-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k v_k(v_{22}v_{11k}-v_{12}v_{12k})\\=&-\frac{2v_{11}}{v_2(v_{11}v_{22}-v_{12}^2)}\sum_k v_k(v_{12}v_{1k}+v_{22}v_{2k})\end{aligned}$$

and

$$\begin{aligned}&\frac{v_{11}}{v_2(v_{11}v_{22}-v_{12}^2)}\sum_k(v_{12}v_{1kk}+v_{22}v_{2kk})\\=&\frac{2v_{11}}{v_2(v_{11}v_{22}-v_{12}^2)}\sum_k v_k(v_{12}v_{1k}+v_{22}v_{2k})-\frac{v_{11}(v_1v_{12}+v_2v_{22})}{v_2(v_{11}v_{22}-v_{12}^2)},\end{aligned}$$

respectively. Thus, (3.5) shows that

$$\begin{aligned}
&-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k v_k(v_{22}v_{11k}-v_{12}v_{12k})\\
&+\frac{v_{11}}{v_2(v_{11}v_{22}-v_{12}^2)}\sum_k(v_{12}v_{1kk}+v_{22}v_{2kk})\\
=&-\frac{v_{11}(v_1v_{12}+v_2v_{22})}{v_2(v_{11}v_{22}-v_{12}^2)}\\
=&-1.
\end{aligned}\tag{3.13}$$

For the quadratic terms in the third derivatives in (3.12), we use (3.8) and the positive definiteness of $\nabla^2 v$ to express v_{11k} in terms of v_{12k}, namely

$$v_{11k}=\frac{v_{12}}{v_{22}}v_{12k}+\frac{v_{11}}{v_2v_{22}}(v_{12}v_{1k}+v_{22}v_{2k}).$$

It follows that

$$v_1v_{11k}+v_2v_{12k}+v_{11}v_{1k}+v_{12}v_{2k}=\frac{v_{11}v_{22}-v_{12}^2}{v_{22}}\left(v_{1k}+\frac{v_2}{v_{11}}v_{12k}\right)\tag{3.14}$$

by (3.4) and

$$\begin{aligned}
&-v_{11k}v_{22k}+v_{12k}^2+\frac{v_{11}}{v_2}(v_{1k}v_{12k}+v_{2k}v_{22k})\\
=&\frac{v_{11}}{v_2v_{22}}(v_{22}v_{12k}-v_{12}v_{22k})\left(v_{1k}+\frac{v_2}{v_{11}}v_{12k}\right).
\end{aligned}\tag{3.15}$$

Combining (3.13)-(3.15) with (3.12), we conclude from (3.6) that

$$\begin{aligned}
\frac{\Delta\phi}{2\phi}=&\frac{1}{\phi v_{22}^2}\sum_k\left(v_{1k}+\frac{v_2}{v_{11}}v_{12k}\right)^2\\
&+\frac{2v_{11}}{v_2v_{22}(v_{11}v_{22}-v_{12}^2)}\sum_k(v_{22}v_{12k}-v_{12}v_{22k})\left(v_{1k}+\frac{v_2}{v_{11}}v_{12k}\right)\\
&-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k(v_{22}v_{1k}^2-v_{12}v_{1k}v_{2k})\\
&-\frac{2\lambda v_{22}}{v_{11}v_{22}-v_{12}^2}-\frac{2(v_1^2v_{22}-v_1v_2v_{12})}{v_{11}v_{22}-v_{12}^2}+3.
\end{aligned}\tag{3.16}$$

We first express the third derivatives in (3.16) in terms of v_{121} and v_{122}. Differentiating (3.1) with respect to x_1 and using (3.3), we obtain

$$v_{111}+v_{221}=0.$$

For differentiation in the x_2 direction, (3.5) leads to

$$v_{112}+v_{222}=\frac{2v_2}{v_{11}}(v_{11}v_{22}-v_{12}^2).$$

Applying (2.7) and substituting the expression for v_1 from (3.4), we rewrite these identities as

$$\begin{aligned} v_{112} &= v_{121} + v_2, \\ v_{111} &= -v_{221} = -v_{122} + \frac{v_2 v_{12}}{v_{11}}, \\ v_{222} &= \frac{2v_2}{v_{11}}(v_{11}v_{22} - v_{12}^2) - v_{121} - v_2. \end{aligned} \tag{3.17}$$

We now determine v_{121} and v_{122} from (3.8). It follows directly from (3.17) and (3.6) that

$$\begin{aligned} v_{12}v_{121} + v_{22}v_{122} &= -\frac{v_{11}v_{12}}{v_2}\big(v_{11} + (1-\phi)v_{22}\big), \\ v_{22}v_{121} - v_{12}v_{122} &= \frac{v_{11}}{v_2}(v_{12}^2 + v_{22}^2 - \phi v_{11}v_{22}). \end{aligned}$$

Since $v_{12}^2 + v_{22}^2 > 0$, the resulting expressions are

$$v_{121} = \frac{v_{11}}{v_2}\left(v_{22} - v_{11} + (1-\phi)\frac{v_{22}(v_{11}v_{22} - v_{12}^2)}{v_{12}^2 + v_{22}^2}\right)$$

and

$$v_{122} = -\frac{v_{11}v_{12}}{v_2}\left(2 - \phi + (1-\phi)\frac{v_{11}v_{22} - v_{12}^2}{v_{12}^2 + v_{22}^2}\right).$$

Taking $k = 1, 2$ in the first two terms on the right-hand side of (3.16), we find

$$\begin{aligned} \frac{1}{v_{22}}\left(v_{11} + \frac{v_2}{v_{11}}v_{121}\right) &= 1 + (1-\phi)\frac{v_{11}v_{22} - v_{12}^2}{v_{12}^2 + v_{22}^2}, \\ \frac{1}{v_{22}}\left(v_{12} + \frac{v_2}{v_{11}}v_{122}\right) &= -(1-\phi)\frac{v_{12}(v_{11} + v_{22})}{v_{12}^2 + v_{22}^2}, \\ v_{22}v_{121} - v_{12}v_{221} &= \frac{v_{11}}{v_2}\big(v_{12}^2 + v_{22}^2 - \phi(v_{11}v_{22} - v_{12}^2)\big) \end{aligned}$$

and

$$v_{22}v_{122} - v_{12}v_{222} = -(1-\phi)\frac{v_{11}v_{12}(v_{11} + v_{22})}{v_2} - \frac{2v_2 v_{12}(v_{11}v_{22} - v_{12}^2)}{v_{11}}.$$

Substituting these identities into (3.16), we arrive at

$$\begin{aligned}\frac{\Delta\phi}{2\phi} = \frac{1}{\phi}\Bigg(&\left(1+(1-\phi)\frac{v_{11}v_{22}-v_{12}^2}{v_{12}^2+v_{22}^2}\right)^2+(1-\phi)^2\frac{v_{12}^2(v_{11}+v_{22})^2}{(v_{12}^2+v_{22}^2)^2}\Bigg)\\ &+\frac{2}{\phi(v_{11}v_{22}-v_{12}^2)}\Bigg(\left(v_{12}^2+v_{22}^2-\phi(v_{11}v_{22}-v_{12}^2)\right)\\ &\qquad\cdot\left(1+(1-\phi)\frac{v_{11}v_{22}-v_{12}^2}{v_{12}^2+v_{22}^2}\right)+(1-\phi)^2\frac{v_{12}^2(v_{11}+v_{22})^2}{v_{12}^2+v_{22}^2}\Bigg)\\ &+4(1-\phi)\frac{v_{12}^2(v_{11}+v_{22})}{v_{12}^2+v_{22}^2}-\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k(v_{22}v_{1k}^2-v_{12}v_{1k}v_{2k})\\ &-\frac{2\lambda v_{22}}{v_{11}v_{22}-v_{12}^2}-2\frac{v_1^2v_{22}-v_1v_2v_{12}}{v_{11}v_{22}-v_{12}^2}+3.\end{aligned} \tag{3.18}$$

Finally, we compute the lower-order terms in (3.18). Direct calculation shows that

$$\begin{aligned}&\left(1+(1-\phi)\frac{v_{11}v_{22}-v_{12}^2}{v_{12}^2+v_{22}^2}\right)^2+(1-\phi)^2\frac{v_{12}^2(v_{11}+v_{22})^2}{(v_{12}^2+v_{22}^2)^2}\\ &=1+2(1-\phi)\frac{v_{11}v_{22}-v_{12}^2}{v_{12}^2+v_{22}^2}+(1-\phi)^2\frac{v_{11}^2+v_{12}^2}{v_{12}^2+v_{22}^2},\end{aligned} \tag{3.19}$$

$$\begin{aligned}&\left(v_{12}^2+v_{22}^2-\phi(v_{11}v_{22}-v_{12}^2)\right)\left(1+(1-\phi)\frac{v_{11}v_{22}-v_{12}^2}{v_{12}^2+v_{22}^2}\right)\\ &+(1-\phi)^2\frac{v_{12}^2(v_{11}+v_{22})^2}{v_{12}^2+v_{22}^2}\\ &=v_{12}^2+v_{22}^2+(1-2\phi)(v_{11}v_{22}-v_{12}^2)+(1-\phi)^2(v_{11}^2+v_{12}^2)\\ &-(1-\phi)\frac{(v_{11}v_{22}-v_{12}^2)^2}{v_{12}^2+v_{22}^2},\end{aligned} \tag{3.20}$$

$$\frac{2}{v_{11}v_{22}-v_{12}^2}\sum_k(v_{22}v_{1k}^2-v_{12}v_{1k}v_{2k})=2v_{11} \tag{3.21}$$

and

$$\begin{aligned}-\frac{2\lambda v_{22}}{v_{11}v_{22}-v_{12}^2}-2\frac{v_1^2v_{22}-v_1v_2v_{12}}{v_{11}v_{22}-v_{12}^2}&=-\frac{2v_{22}(v_{11}+v_{22})}{v_{11}v_{22}-v_{12}^2}+2\phi v_{11}\\ &=2\phi v_{11}-2-2\frac{v_{12}^2+v_{22}^2}{v_{11}v_{22}-v_{12}^2},\end{aligned} \tag{3.22}$$

where we have used (3.1), (3.4) and (3.6). Hence, (3.19)-(3.22) reduce (3.18) to

$$\begin{aligned}\frac{\Delta\phi}{2\phi} &= \frac{1-\phi}{\phi}\left(3+2\frac{v_{12}^2+v_{22}^2}{v_{11}v_{22}-v_{12}^2}-2\phi v_{11}\right)\\ &\quad+\frac{(1-\phi)^2}{\phi}(v_{11}^2+v_{12}^2)\left(\frac{2}{v_{11}v_{22}-v_{12}^2}+\frac{1}{v_{12}^2+v_{22}^2}\right)\\ &\quad+4(1-\phi)\frac{v_{12}^2(v_{11}+v_{22})}{v_{12}^2+v_{22}^2}.\end{aligned}\tag{3.23}$$

Since $\nabla^2 v > 0$ and $0 < \phi < 1$, the last two terms in (3.23) are nonnegative. For the first term, we apply (3.1), (3.4) and (3.6) once more to obtain

$$3+2\frac{v_{12}^2+v_{22}^2}{v_{11}v_{22}-v_{12}^2}-2\phi v_{11} = 1+\frac{2}{v_{11}}\left(\lambda+\phi v_{12}^2+\frac{v_{12}^2(v_{11}+v_{22})}{v_{11}v_{22}-v_{12}^2}\right) > 0.$$

Therefore, $\Delta\phi(x_0) > 0$, contrary to $\Delta\phi(x_0) \le 0$. This completes the proof. □

3.2. **Proof of Theorem 1.1.** We now combine the maximum principle in Proposition 3.1 with the eigenvalue bound $\lambda > \frac{1}{4}$ to rule out Hessian degeneration along the domain deformation from Section 2.5 and prove Theorem 1.1.

Proof of Theorem 1.1. Consider the domain deformation $\{\Omega_t\}_{t\ge 0}$ from Section 2.5. Let u_t be the first Dirichlet eigenfunction on Ω_t with eigenvalue λ_t and set $v_t = -\log u_t$. Denote the curvature of the boundary $\partial\Omega_t$ by κ_t and define ϕ_t by (2.12) with v replaced by v_t. According to Proposition 2.5, there exists $T > 0$ such that $\nabla^2 v_t > 0$ and $\phi_t < 1$ throughout Ω_t whenever $t \ge T$. Let $t_* \in [0, T)$ be the smallest time for which both conditions hold for all $t > t_*$. We first verify that both conditions remain valid at $t = t_*$ and then show that $t_* = 0$.

By continuity, $\nabla^2 v_{t_*} \ge 0$ in Ω_{t_*}. For the computations below we write $v = v_{t_*}$ and $\lambda = \lambda_{t_*}$. Combining $\phi_t < 1$ with (2.12) and letting $t \to t_*$ leads to

$$(v_{11}v_1+v_{12}v_2)^2+(v_{12}v_1+v_{22}v_2)^2 \le (v_{11}v_{22}-v_{12}^2)^2.\tag{3.24}$$

Suppose that $\nabla^2 v$ degenerates at some $x_0 \in \Omega_{t_*}$. Choose orthonormal coordinates at x_0 such that $\nabla^2 v(x_0)$ is diagonal. Since $\nabla^2 v \ge 0$ and (3.1) guarantees $\Delta v > 0$, we may assume $v_{11} > 0$ and $v_{12} = v_{22} = 0$ at x_0. Then, (3.24) forces $v_1 = 0$. All computations below are performed at x_0.

Since v_{22} attains the local minimum 0 at x_0, it follows that $v_{222} = 0$. Differentiating equation (3.1) with respect to x_2, we obtain

$$v_{112} = -v_{222}+2v_1v_{12}+2v_2v_{22} = 0.$$

The commutation formula (2.7) then gives

$$v_{121} = v_{211} = v_{112}-v_2 = -v_2.$$

On the other hand, (3.24) and the positive semi-definiteness of $\nabla^2 v$ provide the estimate

$$|v_{11}v_1 + v_{12}v_2| \leq v_{11}v_{22} - v_{12}^2.$$

Both $v_{11}v_{22} - v_{12}^2$ and $v_{11}v_{22} - v_{12}^2 + v_{11}v_1 + v_{12}v_2$ attain the local minimum 0 at x_0 and hence

$$\nabla(v_{11}v_1 + v_{12}v_2)(x_0) = 0.$$

In particular, the relations $v_{12} = v_1 = 0$ and $v_{121} = -v_2$ simplify the derivative in the x_1 direction to

$$0 = v_2 v_{121} + v_{11}^2 = v_{11}^2 - v_2^2.$$

Furthermore, equation (3.1) can now be written as

$$\lambda = v_{11} - v_2^2 = v_{11}(1 - v_{11}) \leq \frac{1}{4},$$

which contradicts Lemma 2.1. Hence, $\nabla^2 v_{t_*} > 0$ throughout Ω_{t_*}.

Now that ϕ_{t_*} is well defined, letting $t \to t_*$ in the estimate of Proposition 3.1 for $t > t_*$ implies

$$\phi_{t_*} \leq \max_{\partial\Omega_{t_*}} \kappa_{t_*}^{-2} < 1.$$

It follows that both conditions still hold at $t = t_*$.

We now show that $\nabla^2 v_t > 0$ and $\phi_t < 1$ in Ω_t for all $t \geq 0$. Suppose for contradiction that $t_* > 0$. By continuity, both conditions continue to hold on compact subsets of Ω_{t_*} for t sufficiently close to t_*. Near the boundary, Proposition 2.3 and (2.24) ensure that they hold uniformly for t close to t_*. Hence, there exists $\varepsilon \in (0, t_*)$ such that $\nabla^2 v_t > 0$ and $\phi_t < 1$ throughout Ω_t for every $t \in [t_* - \varepsilon, t_*]$. This contradicts the definition of t_*. Thus, $t_* = 0$ and $\nabla^2 v > 0$ in Ω.

It remains to prove the assertion about level sets. Since Ω is strictly horo-convex and v is strictly convex with $v \to +\infty$ at the boundary, v has exactly one critical point. This point is its unique minimum point and the unique maximum point of u. Therefore, each level set Σ_t with $0 < t < \max_\Omega u$ is a smooth closed curve and is strictly convex by (2.4). Furthermore, (2.13) and (3.2) imply

$$\frac{1}{K^2} \leq \phi \leq \max_{\partial\Omega} \kappa^{-2}$$

and hence

$$K \geq \min_{\partial\Omega} \kappa > 1.$$

This completes the proof. □

At the end of this section, we prove Corollary 1.2, where the domain is assumed to be horo-convex rather than strictly horo-convex.

Proof of Corollary 1.2. As in Wei and Xiao [27, Section 4.1], we first approximate Ω by smooth horo-convex domains from which their flow can start. Their Theorem 3.1 together with the dimension reduction in their Section 4.1 ensures strict horo-convexity at positive times. Choosing sufficiently small positive times, we obtain smooth strictly horo-convex domains Ω_ε with $\partial\Omega_\varepsilon \to \partial\Omega$ in C^2.

Let u_ε be the first Dirichlet eigenfunction on Ω_ε with the same normalization as u and set $v_\varepsilon = -\log u_\varepsilon$. Standard elliptic theory ensures that $u_\varepsilon \to u$ and $v_\varepsilon \to v$ locally smoothly in Ω. Theorem 1.1 and the estimates in its proof therefore imply

$$\nabla^2 v \geq 0 \quad \text{in } \Omega$$

and

$$(v_{11}v_1 + v_{12}v_2)^2 + (v_{12}v_1 + v_{22}v_2)^2 \leq (v_{11}v_{22} - v_{12}^2)^2.$$

Since $\lambda > \frac{1}{4}$ by Lemma 2.1, the rank-one degeneration argument in Section 3.2 applies. Hence,

$$\nabla^2 v > 0 \quad \text{in } \Omega.$$

□

4. The three-dimensional case

We turn to the proof of Theorem 1.3 in dimension three. Let u be the first Dirichlet eigenfunction on a bounded, smooth and strictly horo-convex domain $\Omega \subset \mathbb{H}^3$ satisfying (1.1) and set $v = -\log u$. The proof follows the same strategy as in dimension two. Throughout this section, the auxiliary function ϕ is defined in (2.14) and κ_1, κ_2 denote the principal curvatures of $\partial\Omega$. The Latin indices i, j, k, l, m range over $1, 2, 3$ and the Greek index α over $1, 2$.

4.1. A maximum principle.

We first bound ϕ by its boundary value.

Proposition 4.1. *Suppose that $\nabla^2 v > 0$ and $\phi < 1$ in Ω. Then*

$$\phi \leq \max_{\partial\Omega} (\kappa_1\kappa_2)^{-1} < 1. \tag{4.1}$$

Proof. According to (2.25), ϕ extends continuously to $\bar{\Omega}$ with positive boundary values $(\kappa_1\kappa_2)^{-1}$. If (4.1) fails, ϕ reaches its maximum at some $x_0 \in \Omega$. At x_0, $0 < \phi(x_0) < 1$, $\nabla\phi(x_0) = 0$ and $\Delta\phi(x_0) \leq 0$. Since $\phi(x_0) > 0$ and $\nabla^2 v > 0$, we have $\nabla|\nabla v|^2(x_0) \neq 0$. Choose normal orthonormal coordinates at x_0 so that the third coordinate vector points in this direction and the Hessian is diagonal on its orthogonal plane. In these coordinates, $v_{12} = 0$ and the first two components of $\nabla|\nabla v|^2$ vanish. Thus, $v_1 v_{11} + v_3 v_{13} = 0$ and $v_2 v_{22} + v_3 v_{23} = 0$. Since $v_{11} > 0$ and $v_{22} > 0$, these identities imply

$$v_1 = -\frac{v_{13}}{v_{11}} v_3 \quad \text{and} \quad v_2 = -\frac{v_{23}}{v_{22}} v_3. \tag{4.2}$$

The positive definiteness of the Hessian also ensures that

$$v_{33}-\frac{v_{13}^2}{v_{11}}-\frac{v_{23}^2}{v_{22}}=\frac{\det(v_{ij})}{v_{11}v_{22}}>0.$$

By the choice of frame and (4.2),

$$0<v_1v_{13}+v_2v_{23}+v_3v_{33}=v_3\bigg(v_{33}-\frac{v_{13}^2}{v_{11}}-\frac{v_{23}^2}{v_{22}}\bigg),$$

so $v_3>0$ by the preceding inequality.

All calculations below are performed at x_0. The relations (4.2) show that

$$\sum_{i,j}v_iv_jv_{ij}=v_3^2\bigg(v_{33}-\frac{v_{13}^2}{v_{11}}-\frac{v_{23}^2}{v_{22}}\bigg), \tag{4.3}$$

and reduce ϕ to

$$\phi=\frac{v_3^2}{v_{11}v_{22}}. \tag{4.4}$$

Differentiating (2.14) and applying $\nabla\phi(x_0)=0$, we have

$$\begin{aligned}0=\frac{\phi_k}{\phi}&=\frac{\sum_{i,j}v_iv_jv_{ijk}+2\sum_{i,j}v_iv_{ij}v_{jk}}{\sum_{i,j}v_iv_jv_{ij}}-\sum_{i,j}v^{ij}v_{ijk}\\&=2\frac{v_{3k}}{v_3}-\frac{v_{11k}}{v_{11}}-\frac{v_{22k}}{v_{22}}.\end{aligned} \tag{4.5}$$

Here the derivative of the determinant is computed from the identity

$$\frac{(\det(v_{ij}))_k}{\det(v_{ij})}=\sum_{i,j}v^{ij}v_{ijk}.$$

Differentiating (2.14) once more, summing over k and using $\nabla\phi=0$, we arrive at

$$\begin{aligned}\frac{\Delta\phi}{\phi}=&\frac{1}{\sum_{i,j}v_iv_jv_{ij}}\sum_{i,j,k}\big(v_iv_jv_{ijkk}+4v_iv_{jk}v_{ijk}+2v_{ij}v_{ik}v_{jk}+2v_iv_{ij}v_{jkk}\big)\\&-\sum_{i,j,k}v^{ij}v_{ijkk}+\sum_{i,j,l,m,k}v^{ij}v^{lm}\big(v_{ilk}v_{jmk}-v_{ijk}v_{lmk}\big).\end{aligned} \tag{4.6}$$

We first handle the fourth-order derivative terms in (4.6). Taking the derivative of equation (3.1) and invoking (2.7) and (2.6), we obtain

$$\sum_k v_{ikk}=(\Delta v)_i-2v_i=2\sum_k v_kv_{ki}-2v_i, \tag{4.7}$$

and

$$\begin{aligned}(\Delta v)_{ij}&=2\sum_k v_{ki}v_{kj}+2\sum_k v_kv_{kij}\\&=2\sum_k v_{ki}v_{kj}+2\sum_k v_kv_{ijk}+2v_iv_j-2\delta_{ij}\sum_k v_k^2.\end{aligned}$$

Furthermore, (2.8) implies

$$\begin{aligned}\sum_k v_{ijkk} &= (\Delta v)_{ij} - 6v_{ij} + 2\delta_{ij}\Delta v \\ &= 2\sum_k v_k v_{ijk} + 2\sum_k v_{ki}v_{kj} + 2v_i v_j - 6v_{ij} + 2\lambda\delta_{ij}.\end{aligned} \tag{4.8}$$

To simplify the coefficients of the fourth-order derivatives, we use (4.3) and the explicit expressions for the inverse Hessian to write

$$\frac{v_i v_j}{\sum_{l,m} v_l v_m v_{lm}} - v^{ij} = -\frac{\delta_{i1}\delta_{j1}}{v_{11}} - \frac{\delta_{i2}\delta_{j2}}{v_{22}}. \tag{4.9}$$

Together with (4.8), this identity reduces the fourth-order derivative terms in (4.6) to

$$\begin{aligned}\sum_{i,j}\left(\frac{v_i v_j}{\sum_{l,m} v_l v_m v_{lm}} - v^{ij}\right)\sum_k v_{ijkk} &= -\frac{1}{v_{11}}\sum_k v_{11kk} - \frac{1}{v_{22}}\sum_k v_{22kk} \\ &= -2\sum_k v_k\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right) - 2\left(v_{11} + v_{22} + \frac{v_{13}^2}{v_{11}} + \frac{v_{23}^2}{v_{22}}\right) \\ &\quad - 2\left(\frac{v_1^2}{v_{11}} + \frac{v_2^2}{v_{22}}\right) + 12 - 2\lambda\left(\frac{1}{v_{11}} + \frac{1}{v_{22}}\right),\end{aligned}$$

where $v_{12} = 0$ has been used in the last equality. Hence, the preceding identity and (4.7) allow us to rewrite (4.6) as

$$\begin{aligned}\frac{\Delta\phi}{\phi} &= -2\sum_k v_k\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right) - 2\left(v_{11} + v_{22} + \frac{v_{13}^2}{v_{11}} + \frac{v_{23}^2}{v_{22}}\right) \\ &\quad - 2\left(\frac{v_1^2}{v_{11}} + \frac{v_2^2}{v_{22}}\right) + 8 - 2\lambda\left(\frac{1}{v_{11}} + \frac{1}{v_{22}}\right) + \frac{4}{\sum_{i,j} v_i v_j v_{ij}}\sum_k\left(\sum_i v_i v_{ik}\right)^2 \\ &\quad + \frac{1}{\sum_{i,j} v_i v_j v_{ij}}\left(4\sum_{i,j,k} v_i v_{jk} v_{ijk} + 2\sum_{i,j,k} v_{ij} v_{ik} v_{jk}\right) \\ &\quad + \sum_{i,j,l,m,k} v^{ij} v^{lm}\left(v_{ilk} v_{jmk} - v_{ijk} v_{lmk}\right).\end{aligned} \tag{4.10}$$

We next deal with the third-order derivative terms in (4.10) and combine them with the lower-order terms. For the first term and the term containing $(\sum_i v_i v_{ik})^2$, (4.2) and (4.5) imply

$$\begin{aligned}&-2\sum_k v_k\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right) + \frac{4}{\sum_{i,j} v_i v_j v_{ij}}\sum_k\left(\sum_i v_i v_{ik}\right)^2 \\ &= -\frac{4}{v_3}\sum_k v_k v_{3k} + 4\left(v_{33} - \frac{v_{13}^2}{v_{11}} - \frac{v_{23}^2}{v_{22}}\right) = 0.\end{aligned}$$

By (4.9), the last term expands as

$$
\begin{aligned}
&\sum_{i,j,l,m,k} v^{ij}v^{lm}(v_{ilk}v_{jmk}-v_{ijk}v_{lmk})\\
=&\sum_{\alpha,\beta=1}^{2}\sum_{k}\frac{v_{\alpha\beta k}^2}{v_{\alpha\alpha}v_{\beta\beta}}+\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{\alpha=1}^{2}\sum_{k}\frac{1}{v_{\alpha\alpha}}\sum_{i,j}v_iv_jv_{\alpha ik}v_{\alpha jk}\\
&+\frac{1}{\left(\sum_{i,j}v_iv_jv_{ij}\right)^2}\sum_{k}\sum_{i,j,l,m}v_iv_jv_lv_mv_{ilk}v_{jmk}-\sum_{k}\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}+\frac{\sum_{i,j}v_iv_jv_{ijk}}{\sum_{i,j}v_iv_jv_{ij}}\right)^2\\
=&\sum_{k}\left(\frac{v_{11k}^2}{v_{11}^2}+\frac{v_{22k}^2}{v_{22}^2}+\frac{2v_{12k}^2}{v_{11}v_{22}}\right)+\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{\alpha=1}^{2}\sum_{k}\frac{1}{v_{\alpha\alpha}}\left(\sum_{i}v_iv_{\alpha ik}\right)^2\\
&+\frac{1}{\left(\sum_{i,j}v_iv_jv_{ij}\right)^2}\sum_{k}\left(\sum_{i,j}v_iv_jv_{ijk}\right)^2-\sum_{k}\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)^2\\
&-\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{k}\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)\sum_{i,j}v_iv_jv_{ijk}-\frac{1}{\left(\sum_{i,j}v_iv_jv_{ij}\right)^2}\sum_{k}\left(\sum_{i,j}v_iv_jv_{ijk}\right)^2\\
=&\sum_{k}\left(\frac{v_{11k}^2}{v_{11}^2}+\frac{v_{22k}^2}{v_{22}^2}+\frac{2v_{12k}^2}{v_{11}v_{22}}-\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)^2\right)+\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{\alpha=1}^{2}\sum_{k}\frac{1}{v_{\alpha\alpha}}\left(\sum_{i}v_iv_{\alpha ik}\right)^2\\
&-\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{k}\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)\sum_{i,j}v_iv_jv_{ijk}.
\end{aligned}
$$

We combine the final term above with the term containing $\sum_{i,j,k}v_iv_{jk}v_{ijk}$ in (4.10) to obtain

$$
\begin{aligned}
&\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{k}\left(2\sum_{i,j}v_iv_{jk}v_{ijk}-\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)\sum_{i,j}v_iv_jv_{ijk}\right)\\
=&\frac{4}{\sum_{i,j}v_iv_jv_{ij}}\sum_{\alpha=1}^{2}\sum_{k}\frac{1}{v_{\alpha\alpha}}\left(\sum_{i}v_iv_{\alpha ik}\right)\left(\sum_{i}v_{\alpha i}v_{ik}\right),
\end{aligned}
$$

where we have used (4.2) and (4.5). The term involving $\sum_{i,j,k}v_{ij}v_{ik}v_{jk}$ in (4.10) can be rewritten with (4.9) and (4.5) as

$$
\begin{aligned}
\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{i,j,k}v_{ij}v_{ik}v_{jk}&=\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{\alpha=1}^{2}\sum_{k}\frac{1}{v_{\alpha\alpha}}\left(\sum_{i}v_{\alpha i}v_{ik}\right)^2+\frac{2}{v_3^2}\sum_{k}v_{3k}^2\\
&=\frac{2}{\sum_{i,j}v_iv_jv_{ij}}\sum_{\alpha=1}^{2}\sum_{k}\frac{1}{v_{\alpha\alpha}}\left(\sum_{i}v_{\alpha i}v_{ik}\right)^2+\frac{1}{2}\sum_{k}\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)^2.
\end{aligned}
$$

Consequently, (4.10) becomes

$$
\begin{aligned}
\frac{\Delta\phi}{\phi} &= 8 - 2\left(v_{11} + v_{22} + \frac{v_{13}^2}{v_{11}} + \frac{v_{23}^2}{v_{22}}\right) - 2\left(\frac{v_1^2}{v_{11}} + \frac{v_2^2}{v_{22}}\right) - 2\lambda\left(\frac{1}{v_{11}} + \frac{1}{v_{22}}\right) \\
&\quad + \sum_k \left(\frac{v_{11k}^2}{v_{11}^2} + \frac{v_{22k}^2}{v_{22}^2} + \frac{2v_{12k}^2}{v_{11}v_{22}} - \frac{1}{2}\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right)^2\right) \\
&\quad + \frac{2}{\sum_{i,j} v_i v_j v_{ij}} \sum_{\alpha=1}^{2} \sum_k \frac{1}{v_{\alpha\alpha}} \\
&\quad \left(\left(\sum_i v_i v_{\alpha ik}\right)^2 + 2\left(\sum_i v_i v_{\alpha ik}\right)\left(\sum_i v_{\alpha i} v_{ik}\right) + \left(\sum_i v_{\alpha i} v_{ik}\right)^2\right) \\
&= 8 - 2\left(v_{11} + v_{22} + \frac{v_{13}^2}{v_{11}} + \frac{v_{23}^2}{v_{22}}\right) - 2\left(\frac{v_1^2}{v_{11}} + \frac{v_2^2}{v_{22}}\right) - 2\lambda\left(\frac{1}{v_{11}} + \frac{1}{v_{22}}\right) \\
&\quad + \frac{1}{2}\sum_k \left(\frac{v_{11k}}{v_{11}} - \frac{v_{22k}}{v_{22}}\right)^2 + \frac{2}{v_{11}v_{22}} \sum_k v_{12k}^2 \\
&\quad + \frac{2}{\sum_{i,j} v_i v_j v_{ij}} \sum_{\alpha=1}^{2} \sum_k \frac{1}{v_{\alpha\alpha}} \left(\sum_i v_i v_{\alpha ik} + \sum_i v_{\alpha i} v_{ik}\right)^2.
\end{aligned} \tag{4.11}
$$

We continue to estimate the third-order derivative terms in (4.11). Recalling (4.5), we have

$$
v_{22k} = v_{22}\left(\frac{2v_{3k}}{v_3} - \frac{v_{11k}}{v_{11}}\right) \tag{4.12}
$$

and hence

$$
\frac{1}{2}\sum_k \left(\frac{v_{11k}}{v_{11}} - \frac{v_{22k}}{v_{22}}\right)^2 = \frac{2}{v_3^2}\sum_k \left(\frac{v_3 v_{11k}}{v_{11}} - v_{3k}\right)^2. \tag{4.13}
$$

From the commutation formulas (2.6) and (2.7) together with (4.2) and (4.5), we obtain

$$
v_{121} = v_{112} + \frac{v_3 v_{23}}{v_{22}}
$$

and

$$
v_{122} = v_{22}\left(\frac{2v_{13}}{v_3} - \frac{v_{111}}{v_{11}}\right) + \frac{v_3 v_{13}}{v_{11}}.
$$

It follows from these identities and (4.4) that

$$
\begin{aligned}
\frac{2}{v_{11}v_{22}} \sum_k v_{12k}^2 &= \frac{2}{v_3^2}\phi(v_{121}^2 + v_{122}^2 + v_{123}^2) \\
&= \frac{2}{v_3^2}\left(\frac{v_{11}}{v_{22}}\left(\frac{v_3 v_{112}}{v_{11}} + \phi v_{23}\right)^2 + \frac{v_{22}}{v_{11}}\left(\frac{v_3 v_{111}}{v_{11}} - (2+\phi)v_{13}\right)^2 + \phi v_{123}^2\right).
\end{aligned} \tag{4.14}
$$

Since $\nabla^2 v > 0$, we can keep only the terms with $k = \alpha$ in the last term of (4.11). Using (4.3) and applying Young's inequality, we conclude that

$$\begin{aligned}
&\frac{2}{\sum_{i,j} v_i v_j v_{ij}} \sum_{\alpha=1}^{2} \sum_k \frac{1}{v_{\alpha\alpha}} \Big(\sum_i v_i v_{\alpha ik} + \sum_i v_{\alpha i} v_{ik} \Big)^2 \\
&\geq \frac{2}{v_3^2 \big(v_{33} - \frac{v_{13}^2}{v_{11}} - \frac{v_{23}^2}{v_{22}} \big)} \sum_{\alpha=1}^{2} \frac{1}{v_{\alpha\alpha}} \Big(\sum_i v_i v_{\alpha i\alpha} + \sum_i v_{\alpha i}^2 \Big)^2 \\
&\geq \frac{4}{v_3^2} \sum_{\alpha=1}^{2} \Big(\sum_i v_i v_{\alpha i\alpha} + \sum_i v_{\alpha i}^2 \Big) - \frac{2(v_{11}+v_{22})}{v_3^2} \Big(v_{33} - \frac{v_{13}^2}{v_{11}} - \frac{v_{23}^2}{v_{22}} \Big).
\end{aligned} \tag{4.15}$$

To calculate the first term on the right-hand side of (4.15), we write the third-order terms using v_{111}, v_{112} and v_{113}. Applying (4.12) with $k = 2, 3$ and (2.7), we obtain the following identities

$$\begin{aligned}
v_{222} &= v_{22} \Big(\frac{2v_{23}}{v_3} - \frac{v_{112}}{v_{11}} \Big), \\
v_{232} &= v_{22} \Big(\frac{2v_{33}}{v_3} - \frac{v_{113}}{v_{11}} \Big) - v_3,
\end{aligned}$$

and

$$v_{131} = v_{113} - v_3.$$

These identities combined with the expressions for v_{121} and v_{122} lead to

$$\begin{aligned}
&\sum_{\alpha=1}^{2} \Big(\sum_i v_i v_{\alpha i\alpha} + \sum_i v_{\alpha i}^2 \Big) \\
&= v_1(v_{111} + v_{122}) + v_2(v_{121} + v_{222}) + v_3(v_{131} + v_{232}) + v_{11}^2 + v_{22}^2 + v_{13}^2 + v_{23}^2 \\
&= - \frac{v_3 v_{13}}{v_{11}} \Big(v_{111} + v_{22} \Big(\frac{2v_{13}}{v_3} - \frac{v_{111}}{v_{11}} \Big) + \frac{v_3 v_{13}}{v_{11}} \Big) \\
&\quad - \frac{v_3 v_{23}}{v_{22}} \Big(v_{112} + \frac{v_3 v_{23}}{v_{22}} + v_{22} \Big(\frac{2v_{23}}{v_3} - \frac{v_{112}}{v_{11}} \Big) \Big) \\
&\quad + v_3 \Big(v_{113} + v_{22} \Big(\frac{2v_{33}}{v_3} - \frac{v_{113}}{v_{11}} \Big) - 2v_3 \Big) + v_{11}^2 + v_{22}^2 + v_{13}^2 + v_{23}^2 \\
&= - \frac{v_3 v_{13}(v_{11} - v_{22})}{v_{11}^2} v_{111} - \frac{v_3 v_{23}(v_{11} - v_{22})}{v_{11} v_{22}} v_{112} + \frac{v_3(v_{11} - v_{22})}{v_{11}} v_{113} \\
&\quad + 2v_{22}v_{33} + v_{11}^2 + v_{22}^2 - 2v_3^2 + \Big(1 - \frac{v_{22}}{v_{11}}(2 + \phi) \Big) v_{13}^2 - \Big(1 + \frac{v_{11}}{v_{22}} \phi \Big) v_{23}^2,
\end{aligned} \tag{4.16}$$

where (4.2) has been used in the second equality and (4.4) in the last equality.

Combining (4.13)-(4.16) and collecting terms in $\frac{v_3v_{111}}{v_{11}}$, $\frac{v_3v_{112}}{v_{11}}$ and $\frac{v_3v_{113}}{v_{11}}$, we obtain the following estimate from (4.11)

$$
\begin{aligned}
\frac{\Delta\phi}{\phi} &\geq 8-2\left(v_{11}+v_{22}+\frac{v_{13}^2}{v_{11}}+\frac{v_{23}^2}{v_{22}}\right)-2\left(\frac{v_1^2}{v_{11}}+\frac{v_2^2}{v_{22}}\right)-2\lambda\left(\frac{1}{v_{11}}+\frac{1}{v_{22}}\right)\\
&\quad+\frac{2}{v_3^2}\left(\frac{v_{11}+v_{22}}{v_{11}}\left(\frac{v_3v_{111}}{v_{11}}\right)^2+\frac{v_{11}+v_{22}}{v_{22}}\left(\frac{v_3v_{112}}{v_{11}}\right)^2+\left(\frac{v_3v_{113}}{v_{11}}\right)^2\right.\\
&\qquad-\frac{2(2v_{11}+(1+\phi)v_{22})v_{13}}{v_{11}}\frac{v_3v_{111}}{v_{11}}-\frac{2v_{11}(1-\phi)v_{23}}{v_{22}}\frac{v_3v_{112}}{v_{11}}\\
&\qquad+2(v_{11}-v_{22}-v_{33})\frac{v_3v_{113}}{v_{11}}+\phi v_{123}^2+2v_{11}^2+2v_{22}^2+v_{33}^2-4v_3^2\\
&\qquad\left.+(3v_{22}-v_{11})v_{33}+\left(4+\frac{v_{22}}{v_{11}}(1+\phi)^2\right)v_{13}^2+\frac{v_{11}}{v_{22}}(1-\phi)^2v_{23}^2\right)\\
&=8-2\left(v_{11}+v_{22}+\frac{v_{13}^2}{v_{11}}+\frac{v_{23}^2}{v_{22}}\right)-2\left(\frac{v_1^2}{v_{11}}+\frac{v_2^2}{v_{22}}\right)-2\lambda\left(\frac{1}{v_{11}}+\frac{1}{v_{22}}\right)\\
&\quad+\frac{2}{v_3^2}\left(\frac{v_{11}+v_{22}}{v_{11}}\left(\frac{v_3v_{111}}{v_{11}}-\frac{(2v_{11}+(1+\phi)v_{22})v_{13}}{v_{11}+v_{22}}\right)^2\right.\\
&\qquad+\frac{v_{11}+v_{22}}{v_{22}}\left(\frac{v_3v_{112}}{v_{11}}-\frac{v_{11}(1-\phi)v_{23}}{v_{11}+v_{22}}\right)^2\\
&\qquad+\frac{v_{11}v_{22}(1-\phi)^2}{v_{11}+v_{22}}\left(\frac{v_{13}^2}{v_{11}}+\frac{v_{23}^2}{v_{22}}\right)+\left(\frac{v_3v_{113}}{v_{11}}+v_{11}-v_{22}-v_{33}\right)^2\\
&\qquad\left.+\phi v_{123}^2+(v_{11}+v_{22})v_{33}+(v_{11}+v_{22})^2-4v_3^2\right)\\
&\geq\frac{2(v_{11}+v_{22})}{v_3^2}\Delta v-2\left(v_{11}+v_{22}+\frac{v_{13}^2}{v_{11}}+\frac{v_{23}^2}{v_{22}}\right)-2\left(\frac{v_1^2}{v_{11}}+\frac{v_2^2}{v_{22}}\right)\\
&\quad-2\lambda\left(\frac{1}{v_{11}}+\frac{1}{v_{22}}\right),
\end{aligned}
\tag{4.17}
$$

where the last inequality follows from $v_{11}, v_{22} > 0$ and $\phi > 0$.

Finally, equation (3.1) and (4.2) yield

$$\Delta v=\lambda+v_3^2+\frac{v_3^2v_{13}^2}{v_{11}^2}+\frac{v_3^2v_{23}^2}{v_{22}^2}.$$

Substituting this identity into (4.17) and using (4.2) and (4.4), we conclude that

$$
\begin{aligned}
\frac{\Delta\phi}{\phi}&\geq\frac{2\lambda}{v_3^2}(v_{11}+v_{22})-2\lambda\left(\frac{1}{v_{11}}+\frac{1}{v_{22}}\right)\\
&\quad+2\frac{v_{22}v_{13}^2}{v_{11}^2}+2\frac{v_{11}v_{23}^2}{v_{22}^2}-2\frac{v_3^2v_{13}^2}{v_{11}^3}-2\frac{v_3^2v_{23}^2}{v_{22}^3}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{2\lambda}{v_3^2}(v_{11}+v_{22})(1-\phi) + 2\frac{v_{13}^2}{v_{11}^2}\left(v_{22}-\frac{v_3^2}{v_{11}}\right) + 2\frac{v_{23}^2}{v_{22}^2}\left(v_{11}-\frac{v_3^2}{v_{22}}\right)\\
&= 2(1-\phi)\left(\frac{\lambda(v_{11}+v_{22})}{v_3^2} + \frac{v_{22}v_{13}^2}{v_{11}^2} + \frac{v_{11}v_{23}^2}{v_{22}^2}\right)\\
&> 0,
\end{aligned}
$$

since $\nabla^2 v > 0$, $0 < \phi < 1$ and $\lambda > 0$. This contradicts $\Delta\phi(x_0) \le 0$ and completes the proof. □

4.2. **Proof of Theorem 1.3 for $n = 3$.** With Proposition 4.1 established, we turn to the proof of Theorem 1.3 for $n = 3$. We use the domain deformation as in Section 3.2 and show that neither of the two possible types of Hessian degeneration can occur.

Proof. Let $\{\Omega_t\}_{t\ge 0}$ and v_t be as in Section 2.5. Define ϕ_t by (2.14) with v replaced by v_t. Denote the first Dirichlet eigenvalue of Ω_t by λ_t and its boundary principal curvatures by $\kappa_{1,t}$, $\kappa_{2,t}$. Choose $T > 0$ as in Proposition 2.5 so that $\nabla^2 v_t > 0$ and $\phi_t < 1$ in Ω_t whenever $t \ge T$. We take $t_* \in [0, T)$ to be the smallest time such that both conditions hold for every $t > t_*$. As in the proof of Theorem 1.1, we complete the argument by verifying both conditions at $t = t_*$ and showing that $t_* = 0$.

For the following calculations, write $v = v_{t_*}$ and $\lambda = \lambda_{t_*}$. By continuity, $\nabla^2 v \ge 0$. Taking $t \to t_*$ in (4.1), we obtain

$$
\sum_{i,j} v_i v_j v_{ij} \le \left(\max_{\partial\Omega_{t_*}} (\kappa_{1,t_*}\kappa_{2,t_*})^{-1}\right) \det(v_{ij}). \tag{4.18}
$$

Assume that $\nabla^2 v$ is degenerate at some $x_0 \in \Omega_{t_*}$. Since $\nabla^2 v \ge 0$ and equation (3.1) gives $\Delta v > 0$, its rank at x_0 is either two or one. Moreover, (4.18) shows that both $\det(v_{ij})$ and

$$
\left(\max_{\partial\Omega_{t_*}} (\kappa_{1,t_*}\kappa_{2,t_*})^{-1}\right) \det(v_{ij}) - \sum_{i,j} v_i v_j v_{ij}
$$

attain their minimum value 0 at x_0. Their Laplacians are therefore nonnegative there.

Choose orthonormal coordinates at x_0 such that $\nabla^2 v(x_0)$ is diagonal. All calculations below are performed at x_0.

Case 1. $\nabla^2 v$ has rank two. We may assume that $v_{11} > 0$, $v_{22} > 0$ and $v_{33} = 0$. Then, (4.18) forces $v_1 = v_2 = 0$. To calculate $\Delta \det(v_{ij})$, we note that v_{33} attains the local minimum 0 at x_0. Thus, $v_{33k} = 0$. Differentiating (3.1) with respect to x_3 yields

$$
v_{113} + v_{223} = -v_{333} + 2\sum_i v_i v_{i3} = 0.
$$

The commutation formulas (2.6) and (2.7) therefore imply

$$
\begin{aligned}
v_{131} &= v_{113} - v_3,\\
v_{232} &= -v_{113} - v_3,
\end{aligned}
$$

$$v_{133} = v_{233} = 0,$$

and

$$v_{132} = v_{231} = v_{123}.$$

Moreover, (4.8) reduces to

$$\sum_k v_{33kk} = 2\lambda + 2v_3^2 = 2(v_{11} + v_{22}).$$

It follows that

$$\begin{aligned} \Delta \det(v_{ij}) &= v_{11}v_{22}\sum_k v_{33kk} - 2v_{22}\sum_k v_{13k}^2 - 2v_{11}\sum_k v_{23k}^2 \\ &= 2v_{11}v_{22}(v_{11}+v_{22}) - 2v_{22}\big((v_{113}-v_3)^2 + v_{123}^2\big) \\ &\quad - 2v_{11}\big((v_{113}+v_3)^2 + v_{123}^2\big). \end{aligned} \tag{4.19}$$

Since $\sum_i v_i v_{ij} = 0$ for every j, we also calculate

$$\begin{aligned} \Delta\bigg(\sum_{i,j} v_i v_j v_{ij}\bigg) &= \sum_{i,j} v_i v_j \sum_k v_{ijkk} + 4\sum_{i,j,k} v_i v_{jk} v_{ijk} + 2\sum_{i,j,k} v_{ik}v_{jk}v_{ij} \\ &\quad + 4\sum_j \bigg(\sum_i v_i v_{ij}\bigg)^2 - 4\sum_{i,j} v_i v_j v_{ij} \\ &= v_3^2 \sum_k v_{33kk} + 4v_3(v_{11}v_{131} + v_{22}v_{232}) + 2(v_{11}^3 + v_{22}^3) \\ &= -2v_3^2(v_{11}+v_{22}) + 4v_3(v_{11}-v_{22})v_{113} + 2(v_{11}^3 + v_{22}^3). \end{aligned}$$

Thus,

$$\begin{aligned} &\Delta\bigg(\det(v_{ij}) - \sum_{i,j} v_i v_j v_{ij}\bigg) \\ &= -2(v_{11}+v_{22})(v_{113}^2 + v_{123}^2) - 8v_3(v_{11}-v_{22})v_{113} - 2(v_{11}+v_{22})(v_{11}-v_{22})^2 \\ &= -2(v_{11}+v_{22})\bigg(\bigg(v_{113} + \frac{2v_3(v_{11}-v_{22})}{v_{11}+v_{22}}\bigg)^2 + v_{123}^2\bigg) \\ &\quad + \frac{2(v_{11}-v_{22})^2}{v_{11}+v_{22}}\big(4v_3^2 - (v_{11}+v_{22})^2\big). \end{aligned} \tag{4.20}$$

Recalling that $\lambda > 1$ by Lemma 2.1, we rewrite (3.1) as

$$\lambda = v_{11} + v_{22} - v_3^2$$

and arrive at

$$4v_3^2 - (v_{11}+v_{22})^2 = -(v_{11}+v_{22}-2)^2 - 4(\lambda - 1) < 0.$$

Hence, the right-hand side of (4.20) is nonpositive and the two minimum conditions yield

$$\begin{aligned}0 &\le \Delta\bigg(\Big(\max_{\partial\Omega_{t_*}}(\kappa_{1,t_*}\kappa_{2,t_*})^{-1}\Big)\det(v_{ij}) - \sum_{i,j} v_i v_j v_{ij}\bigg)\\ &= \Delta\bigg(\det(v_{ij}) - \sum_{i,j} v_i v_j v_{ij}\bigg) - \Big(1 - \max_{\partial\Omega_{t_*}}(\kappa_{1,t_*}\kappa_{2,t_*})^{-1}\Big)\Delta\det(v_{ij})\\ &\le -\Big(1 - \max_{\partial\Omega_{t_*}}(\kappa_{1,t_*}\kappa_{2,t_*})^{-1}\Big)\Delta\det(v_{ij}) \le 0.\end{aligned}$$

By strict horo-convexity, $1 - \max_{\partial\Omega_{t_*}}(\kappa_{1,t_*}\kappa_{2,t_*})^{-1} > 0$. Hence,

$$\Delta\det(v_{ij}) = 0 \quad \text{and } \Delta\bigg(\det(v_{ij}) - \sum_{i,j} v_i v_j v_{ij}\bigg) = 0.$$

It follows from (4.20) that $v_{11} = v_{22}$ and $v_{113} = v_{123} = 0$. Relation (4.19) then reduces to $0 = 4v_{11}(v_{11}^2 - v_3^2)$. Since $v_{11} > 0$, this gives $v_3^2 = v_{11}^2$. Therefore,

$$\lambda = 2v_{11} - v_{11}^2 \le 1,$$

in contradiction with Lemma 2.1.

Case 2. $\nabla^2 v$ has rank one. Take $v_{11} > 0$ and $v_{22} = v_{33} = 0$. Then, $v_1 = 0$ by (4.18). As in Case 1, we first calculate $\Delta\det(v_{ij})$. Here the nonnegative functions v_{22}, v_{33}, $v_{22} + v_{33} + 2v_{23}$ and $v_{22} + v_{33} - 2v_{23}$ all vanish at x_0. Thus, $v_{22k} = v_{33k} = v_{23k} = 0$. Since $\sum_i v_i v_{ik} = 0$ for every k, differentiation of (3.1) implies

$$v_{11k} = -v_{22k} - v_{33k} = 0.$$

It follows from (2.7) that $v_{211} = -v_2$ and $v_{311} = -v_3$. Consequently,

$$\Delta\det(v_{ij}) = 2v_{11}\sum_k (v_{22k}v_{33k} - v_{23k}^2) = 0. \tag{4.21}$$

For $i, j \in \{2, 3\}$, all v_{ijk} vanish and (4.8) becomes

$$\sum_k v_{ijkk} = 2v_i v_j + 2\lambda\delta_{ij}.$$

We therefore obtain

$$\begin{aligned}\Delta\bigg(\sum_{i,j} v_i v_j v_{ij}\bigg) &= \sum_{i,j=2}^{3} v_i v_j \sum_k v_{ijkk} + 4v_{11}(v_2 v_{211} + v_3 v_{311}) + 2v_{11}^3\\ &= 2(v_2^2 + v_3^2)^2 + 2\lambda(v_2^2 + v_3^2) - 4v_{11}(v_2^2 + v_3^2) + 2v_{11}^3\\ &= 2v_{11}\big(v_{11}^2 - (v_2^2 + v_3^2)\big),\end{aligned}$$

where the last equality uses $\lambda = v_{11} - v_2^2 - v_3^2$. By strict horo-convexity and (4.18), the function $\det(v_{ij}) - \sum_{i,j} v_i v_j v_{ij}$ also attains its minimum value 0 at x_0. Together with (4.21), this leads to

$$2v_{11}\big(v_{11}^2 - (v_2^2 + v_3^2)\big) \le 0.$$

As $v_{11} > 0$, we conclude that $v_2^2 + v_3^2 \geq v_{11}^2$ and therefore

$$\lambda = v_{11} - v_2^2 - v_3^2 \leq v_{11} - v_{11}^2 \leq \frac{1}{4},$$

again contradicting Lemma 2.1.

We have now proved that $\nabla^2 v_{t_*} > 0$ in Ω_{t_*}, so ϕ_{t_*} is well defined. By (4.18),

$$\phi_{t_*} \leq \max_{\partial\Omega_{t_*}} (\kappa_{1,t_*}\kappa_{2,t_*})^{-1} < 1$$

and hence both conditions hold at $t = t_*$.

As in the proof of Theorem 1.1, continuity, the boundary convexity estimate in Proposition 2.3 and the boundary behavior of ϕ_t stated in (2.25) ensure that both conditions remain valid for times slightly below t_* if $t_* > 0$. This contradicts the definition of t_*. Hence, $t_* = 0$ and $\nabla^2 v > 0$ in Ω.

Finally, the argument at the end of Section 3.2 shows that u has a unique critical point, its maximum point. Every Σ_t with $0 < t < \max_\Omega u$ is a smooth closed surface. Its strict convexity follows from (2.4). Moreover, (2.15) and (4.1) show that

$$\frac{1}{K} \leq \phi \leq \max_{\partial\Omega} (\kappa_1\kappa_2)^{-1}.$$

Equivalently,

$$K \geq \min_{\partial\Omega} (\kappa_1\kappa_2) > 1.$$

This completes the proof in dimension three. □

5. The higher-dimensional case

In this section, we prove Theorem 1.3 for $n \geq 4$. Let u be the first Dirichlet eigenfunction satisfying (1.1) on a bounded, smooth and strictly horo-convex domain $\Omega \subset \mathbb{H}^n$. The proof relies on a minimum principle for $e^{-v}(1-\phi)$, where $v = -\log u$ and ϕ is the auxiliary function defined in (2.17). We then follow the same domain deformation strategy as in dimensions two and three.

Throughout this section, Latin indices i, j, k, l, m, p, q range from 1 to n and a from 1 to 2, while Greek indices α, β range from 3 to $n-1$.

5.1. **A minimum principle.**

The following proposition will provide a uniform upper bound for ϕ along the domain deformation.

Proposition 5.1. *Suppose that $\nabla^2 v > 0$ and $\phi < 1$ in Ω. Then, $e^{-v}(1-\phi)$ has no local minimum in Ω.*

Proof. Suppose to the contrary that $e^{-v}(1-\phi)$ attains a local minimum at some $x_0 \in \Omega$. Then, there exists a neighborhood $\mathcal{O} \subset \Omega$ of x_0 such that

$$e^{-v(x)}\big(1-\phi(x)\big) \geq e^{-v(x_0)}\big(1-\phi(x_0)\big) \quad \text{for all } x \in \mathcal{O}.$$

If $\nabla v(x_0) = 0$, then $\phi(x_0) = 0$, while $\nabla^2 v > 0$ implies that v has a strict local minimum at x_0. Taking $\mathcal{O}$ sufficiently small, we therefore have

$$e^{-v(x)}\big(1 - \phi(x)\big) \leq e^{-v(x)} < e^{-v(x_0)} \quad \text{for } x \in \mathcal{O} \setminus \{x_0\},$$

which is a contradiction. Hence, $\nabla v(x_0) \neq 0$ and $0 < \phi(x_0) < 1$.

We now choose suitable coordinates at x_0 to simplify the subsequent calculations. We first take an orthonormal basis whose last vector is $\nabla v/|\nabla v|$ and rotate the tangent directions to diagonalize the tangential submatrix $(v^{ij})_{1\leq i,j\leq n-1}$ of the inverse Hessian. Choose unit eigenvectors corresponding to the two largest eigenvalues as the first two basis vectors ξ, η. This pair attains the maximum in (2.17). With this choice, the tangential components of $(\nabla^2 v)^{-1}\xi$ and $(\nabla^2 v)^{-1}\eta$ are parallel to ξ, η, respectively, while their normal components are parallel to ∇v. Thus, both vectors belong to the three-dimensional space spanned by ξ, η and ∇v. The gradient does not belong to the plane spanned by $(\nabla^2 v)^{-1}\xi$ and $(\nabla^2 v)^{-1}\eta$. Otherwise, $(\nabla^2 v)\nabla v$ would be a linear combination of ξ, η and hence orthogonal to ∇v, contradicting the positive definiteness of the Hessian and $\nabla v(x_0) \neq 0$. Then, we choose an orthonormal basis of this plane as the first two coordinate directions and take the last coordinate direction along the component of ∇v orthogonal to it. These three directions span the same space as ξ, η and ∇v. Taking the remaining coordinate directions in its orthogonal complement, we obtain orthonormal coordinates at x_0 such that

$$v_n > 0 \quad \text{and } v_\alpha = 0. \tag{5.1}$$

Moreover, the first two coordinate directions are linear combinations of $(\nabla^2 v)^{-1}\xi$ and $(\nabla^2 v)^{-1}\eta$. Applying the Hessian to either direction therefore yields a linear combination of ξ, η. Since the remaining coordinate directions are orthogonal to ξ, η, we have

$$v_{1\alpha} = 0 \quad \text{and } v_{2\alpha} = 0. \tag{5.2}$$

Finally, we rotate the first two coordinate directions and the remaining $n-3$ coordinate directions separately to diagonalize the corresponding Hessian submatrices. The conditions (5.1) and (5.2) remain unchanged and the rotated frame also satisfies

$$v_{12} = 0 \quad \text{and } v_{\alpha\beta} = 0 \ (\alpha \neq \beta). \tag{5.3}$$

All calculations below are performed at x_0 and the coordinate directions are extended by parallel translation along radial geodesics. We first compute the value of ϕ. The vectors with components v_{i1} and v_{i2} span the maximizing tangent plane and are therefore orthogonal to ∇v. Using (5.1) and (5.3), we obtain

$$v_1 = -\frac{v_{1n}}{v_{11}} v_n \quad \text{and } v_2 = -\frac{v_{2n}}{v_{22}} v_n. \tag{5.4}$$

From (5.1)-(5.4), the definition (2.17) takes the form

$$\phi = \frac{v_n^2}{v_{11} v_{22}}. \tag{5.5}$$

To differentiate ϕ, we use the construction described in Section 2.3. At x_0, the tangential projections of the first two coordinate directions span the maximizing tangent plane. Hence, the smooth function in (2.18) agrees with the original auxiliary function (2.17) at x_0 and is bounded above by it nearby. We continue to denote this smooth function by ϕ. Then, $e^{-v}(1-\phi)$ still attains a local minimum at x_0, so

$$\phi_k = -(1-\phi)v_k. \tag{5.6}$$

Moreover,

$$\Delta\big(e^{-v}(1-\phi)\big) \geq 0. \tag{5.7}$$

Differentiating (2.18) using

$$(v^{ij})_k = -\sum_{l,m} v^{il}v^{jm}v_{lmk}$$

and simplifying by (5.1)-(5.4), we obtain

$$\begin{aligned}\phi_k &= \frac{2v_n}{v_{11}v_{22}}v_{nk} - \frac{v_n^2}{v_{11}^2 v_{22}}v_{11k} - \frac{v_n^2}{v_{11}v_{22}^2}v_{22k}\\ &= \phi\left(\frac{2v_{nk}}{v_n} - \frac{v_{11k}}{v_{11}} - \frac{v_{22k}}{v_{22}}\right).\end{aligned} \tag{5.8}$$

For the second derivatives, we also use the identity

$$(v^{ij})_{kk} = 2\sum_{l,m,p,q} v^{il}v^{mp}v^{qj}v_{lmk}v_{pqk} - \sum_{l,m} v^{il}v^{mj}v_{lmkk}.$$

Differentiating (2.18) once more and summing over k, we find

$$\begin{aligned}\frac{\Delta\phi}{\phi} = {}& \frac{2}{v_n^2}\sum_{i=3}^{n}\sum_k (v_{ik}^2 + v_i v_{ikk}) - \frac{4}{v_n^2}\sum_k v_{nk}^2 - \frac{1}{v_{11}}\sum_k v_{11kk} - \frac{1}{v_{22}}\sum_k v_{22kk}\\ &+ \sum_k\left(\frac{v_{11k}^2}{v_{11}^2} + \frac{v_{22k}^2}{v_{22}^2} + \frac{2v_{12k}^2}{v_{11}v_{22}}\right) + \frac{|\nabla\phi|^2}{\phi^2}\\ &+ 2\sum_{a=1}^{2}\sum_k \frac{1}{v_{aa}}\left(\frac{v^{nn}}{v_n^2}\left(\sum_i v_i v_{aik} + \sum_i v_{ai}v_{ik}\right)^2\right.\\ &\qquad + \frac{2}{v_n}\sum_{\alpha=3}^{n-1} v^{n\alpha}\left(\sum_i v_i v_{aik} + \sum_i v_{ai}v_{ik}\right)\left(v_{a\alpha k} - \frac{v_{an}v_{\alpha k}}{v_n}\right)\\ &\qquad \left. + \sum_{\alpha,\beta=3}^{n-1} v^{\alpha\beta}\left(v_{a\alpha k} - \frac{v_{an}v_{\alpha k}}{v_n}\right)\left(v_{a\beta k} - \frac{v_{an}v_{\beta k}}{v_n}\right)\right).\end{aligned} \tag{5.9}$$

To evaluate the fourth-order derivatives in (5.9), we first differentiate equation (3.1) and employ the commutation formula (2.7) to obtain

$$\sum_k v_{ikk} = (\Delta v)_i - (n-1)v_i = 2\sum_k v_k v_{ik} - (n-1)v_i.$$

The second derivatives of (3.1) can be written as

$$\begin{aligned}(\Delta v)_{ij} &= 2\sum_k v_{ki}v_{kj} + 2\sum_k v_k v_{kij}\\ &= 2\sum_k v_{ki}v_{kj} + 2\sum_k v_k v_{ijk} + 2v_i v_j - 2\delta_{ij}\sum_k v_k^2,\end{aligned}$$

where we have used (2.6) in the last equality. It follows from (2.8) that

$$\begin{aligned}\sum_k v_{ijkk} &= (\Delta v)_{ij} - 2nv_{ij} + 2\delta_{ij}\Delta v\\ &= 2\sum_k v_k v_{ijk} + 2\sum_k v_{ki}v_{kj} + 2v_i v_j - 2nv_{ij} + 2\lambda\delta_{ij}.\end{aligned} \tag{5.10}$$

Using (5.2), (5.3) and (5.10), we calculate

$$\begin{aligned}&-\frac{1}{v_{11}}\sum_k v_{11kk} - \frac{1}{v_{22}}\sum_k v_{22kk}\\ =&-2\sum_k v_k\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right) - 2\left(v_{11} + v_{22} + \frac{v_{1n}^2}{v_{11}} + \frac{v_{2n}^2}{v_{22}}\right)\\ &-2\left(\frac{v_1^2}{v_{11}} + \frac{v_2^2}{v_{22}}\right) + 4n - 2\lambda\left(\frac{1}{v_{11}} + \frac{1}{v_{22}}\right).\end{aligned}$$

Note that

$$\sum_{i=3}^n\sum_k v_{ik}^2 - 2\sum_k v_{nk}^2 = \sum_{\alpha=3}^{n-1}(v_{\alpha\alpha}^2 + v_{n\alpha}^2) - \sum_k v_{nk}^2.$$

Thus,

$$
\begin{aligned}
\frac{\Delta\phi}{\phi} &= 4n - 2\left(v_{11} + v_{22} + \frac{v_{1n}^2}{v_{11}} + \frac{v_{2n}^2}{v_{22}}\right) - 2\left(\frac{v_1^2}{v_{11}} + \frac{v_2^2}{v_{22}}\right) - 2\lambda\left(\frac{1}{v_{11}} + \frac{1}{v_{22}}\right) \\
&\quad + \frac{2}{v_n^2}\sum_{i=3}^{n}\sum_{k} v_i v_{ikk} - 2\sum_{k} v_k\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right) \\
&\quad + \sum_{k}\left(\frac{v_{11k}^2}{v_{11}^2} + \frac{v_{22k}^2}{v_{22}^2} + \frac{2v_{12k}^2}{v_{11}v_{22}}\right) + \frac{|\nabla\phi|^2}{\phi^2} \\
&\quad + \frac{2}{v_n^2}\sum_{\alpha=3}^{n-1}(v_{\alpha\alpha}^2 + v_{n\alpha}^2) - \frac{2}{v_n^2}\sum_{k} v_{nk}^2 \\
&\quad + 2\sum_{a=1}^{2}\sum_{k}\frac{1}{v_{aa}}\left(\frac{v^{nn}}{v_n^2}\left(\sum_i v_i v_{aik} + \sum_i v_{ai}v_{ik}\right)^2\right. \\
&\qquad\qquad + \frac{2}{v_n}\sum_{\alpha=3}^{n-1} v^{n\alpha}\left(\sum_i v_i v_{aik} + \sum_i v_{ai}v_{ik}\right)\left(v_{a\alpha k} - \frac{v_{an}v_{\alpha k}}{v_n}\right) \\
&\qquad\qquad \left. + \sum_{\alpha,\beta=3}^{n-1} v^{\alpha\beta}\left(v_{a\alpha k} - \frac{v_{an}v_{\alpha k}}{v_n}\right)\left(v_{a\beta k} - \frac{v_{an}v_{\beta k}}{v_n}\right)\right).
\end{aligned} \tag{5.11}
$$

We next handle the third-order derivative terms. For the linear terms in the third-order derivatives, we have

$$
\begin{aligned}
&\frac{2}{v_n^2}\sum_{i=3}^{n}\sum_{k} v_i v_{ikk} - 2\sum_{k} v_k\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right) \\
&= \frac{4}{v_n}\sum_{k} v_k v_{nk} - 2(n-1) - 2\sum_{k} v_k\left(\frac{v_{11k}}{v_{11}} + \frac{v_{22k}}{v_{22}}\right) \\
&= 2\sum_{k} v_k\frac{\phi_k}{\phi} - 2(n-1),
\end{aligned} \tag{5.12}
$$

where the last equality follows from (5.8). For the quadratic terms, we first note from $v_{12} = 0$ and the positive definiteness of $\nabla^2 v$ that

$$
v_{nn} - \frac{v_{1n}^2}{v_{11}} - \frac{v_{2n}^2}{v_{22}} > 0. \tag{5.13}
$$

Using (5.2) and (5.3), the identities $\sum_j v_{ij}v^{jk} = \delta_{ik}$ imply

$$
v^{n\alpha} = -\frac{\sum_{\beta=3}^{n-1} v^{\alpha\beta}v_{\beta n}}{v_{nn} - \frac{v_{1n}^2}{v_{11}} - \frac{v_{2n}^2}{v_{22}}}
$$

and

$$v^{nn} = \frac{1}{v_{nn} - \frac{v_{1n}^2}{v_{11}} - \frac{v_{2n}^2}{v_{22}}} + \frac{\sum_{\alpha,\beta=3}^{n-1} v^{\alpha\beta} v_{\alpha n} v_{\beta n}}{\left(v_{nn} - \frac{v_{1n}^2}{v_{11}} - \frac{v_{2n}^2}{v_{22}}\right)^2}.$$

Consequently, the last term in (5.11) can be written as

$$\begin{aligned}
&2\sum_{a=1}^{2}\sum_{k}\frac{1}{v_{aa}}\Bigg(\frac{v^{nn}}{v_n^2}\Big(\sum_i v_i v_{aik} + \sum_i v_{ai} v_{ik}\Big)^2 \\
&\qquad + \frac{2}{v_n}\sum_{\alpha=3}^{n-1} v^{n\alpha}\Big(\sum_i v_i v_{aik} + \sum_i v_{ai} v_{ik}\Big)\Big(v_{a\alpha k} - \frac{v_{an}v_{\alpha k}}{v_n}\Big) \\
&\qquad + \sum_{\alpha,\beta=3}^{n-1} v^{\alpha\beta}\Big(v_{a\alpha k} - \frac{v_{an}v_{\alpha k}}{v_n}\Big)\Big(v_{a\beta k} - \frac{v_{an}v_{\beta k}}{v_n}\Big)\Bigg) \\
&= \frac{2}{v_n^2\big(v_{nn} - \frac{v_{1n}^2}{v_{11}} - \frac{v_{2n}^2}{v_{22}}\big)}\sum_{a=1}^{2}\sum_{k}\frac{1}{v_{aa}}\Big(\sum_i v_i v_{aik} + \sum_i v_{ai} v_{ik}\Big)^2 \\
&\quad + 2\sum_{a=1}^{2}\sum_{k}\frac{1}{v_{aa}}\sum_{\alpha,\beta=3}^{n-1} v^{\alpha\beta} \\
&\quad \cdot \Big(v_{a\alpha k} - \frac{v_{an}v_{\alpha k}}{v_n} - \frac{v_{\alpha n}}{v_n\big(v_{nn} - \frac{v_{1n}^2}{v_{11}} - \frac{v_{2n}^2}{v_{22}}\big)}\Big(\sum_i v_i v_{aik} + \sum_i v_{ai} v_{ik}\Big)\Big) \\
&\quad \cdot \Big(v_{a\beta k} - \frac{v_{an}v_{\beta k}}{v_n} - \frac{v_{\beta n}}{v_n\big(v_{nn} - \frac{v_{1n}^2}{v_{11}} - \frac{v_{2n}^2}{v_{22}}\big)}\Big(\sum_i v_i v_{aik} + \sum_i v_{ai} v_{ik}\Big)\Big).
\end{aligned} \tag{5.14}$$

The second term on the right-hand side is nonnegative, since $(v^{\alpha\beta})_{\alpha,\beta=3}^{n-1}$ is positive definite. For the terms involving v_{11k}^2, v_{22k}^2, v_{nk}^2 and $|\nabla\phi|^2$, we apply (5.4), (5.6) and (5.8) to write

$$
\begin{aligned}
&\sum_k \left(\frac{v_{11k}^2}{v_{11}^2}+\frac{v_{22k}^2}{v_{22}^2}\right)+\frac{|\nabla\phi|^2}{\phi^2}-\frac{2}{v_n^2}\sum_k v_{nk}^2\\
&=2\sum_k \left(\frac{v_{11k}^2}{v_{11}^2}+\frac{v_{22k}^2}{v_{22}^2}\right)+\frac{2}{v_{11}v_{22}}\sum_k v_{11k}v_{22k}\\
&\quad-\frac{4}{v_n}\sum_k v_{nk}\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)+\frac{2}{v_n^2}\sum_k v_{nk}^2\\
&=\frac{1}{2}\sum_k \left(\frac{v_{11k}}{v_{11}}-\frac{v_{22k}}{v_{22}}\right)^2+\frac{3}{2}\sum_k \left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)^2\\
&\quad-\frac{4}{v_n}\sum_k v_{nk}\left(\frac{v_{11k}}{v_{11}}+\frac{v_{22k}}{v_{22}}\right)+\frac{2}{v_n^2}\sum_k v_{nk}^2\\
&=\frac{1}{2}\sum_k \left(\frac{v_{11k}}{v_{11}}-\frac{v_{22k}}{v_{22}}\right)^2+\frac{3}{2}\sum_k \left(\frac{2v_{nk}}{3v_n}+\frac{1-\phi}{\phi}v_k\right)^2-\frac{2}{3v_n^2}\sum_k v_{nk}^2\\
&=\frac{1}{2}\sum_k \left(\frac{v_{11k}}{v_{11}}-\frac{v_{22k}}{v_{22}}\right)^2+\frac{2(1-\phi)}{\phi}\left(v_{nn}-\frac{v_{1n}^2}{v_{11}}-\frac{v_{2n}^2}{v_{22}}\right)+\frac{3(1-\phi)^2}{2\phi^2}|\nabla v|^2.
\end{aligned}
\tag{5.15}
$$

The last two terms are nonnegative by (5.13) and $0<\phi(x_0)<1$. Substituting (5.12)-(5.15) into (5.11) and dropping the nonnegative terms including those containing $v_{n\alpha}^2$, we obtain

$$
\begin{aligned}
\frac{\Delta\phi-2\sum_k v_k\phi_k}{\phi}\ge\ & 2n+2-2\left(v_{11}+v_{22}+\frac{v_{1n}^2}{v_{11}}+\frac{v_{2n}^2}{v_{22}}\right)-2\left(\frac{v_1^2}{v_{11}}+\frac{v_2^2}{v_{22}}\right)\\
&-2\lambda\left(\frac{1}{v_{11}}+\frac{1}{v_{22}}\right)+\frac{1}{2}\sum_k\left(\frac{v_{11k}}{v_{11}}-\frac{v_{22k}}{v_{22}}\right)^2+\frac{2}{v_{11}v_{22}}\sum_k v_{12k}^2\\
&+\frac{2}{v_n^2}\sum_{\alpha=3}^{n-1}v_{\alpha\alpha}^2\\
&+\frac{2}{v_n^2\left(v_{nn}-\frac{v_{1n}^2}{v_{11}}-\frac{v_{2n}^2}{v_{22}}\right)}\sum_{a=1}^{2}\sum_k\frac{1}{v_{aa}}\left(\sum_i v_i v_{aik}+\sum_i v_{ai}v_{ik}\right)^2.
\end{aligned}
\tag{5.16}
$$

We further estimate the third-order derivative terms in (5.16) as in the three-dimensional proof. The ingredient is to collect the terms involving $\frac{v_n v_{111}}{v_{11}}$, $\frac{v_n v_{112}}{v_{11}}$ and $\frac{v_n v_{11n}}{v_{11}}$. By (5.6) and (5.8), we express v_{22k} in terms of v_{11k} as

$$
v_{22k}=v_{22}\left(\frac{2v_{nk}}{v_n}-\frac{v_{11k}}{v_{11}}+\frac{1-\phi}{\phi}v_k\right).
\tag{5.17}
$$

Thus, we retain only the terms with $k = 1, 2, n$ and use (5.4) and (5.5) to obtain

$$\begin{aligned}\frac{1}{2}\sum_k\left(\frac{v_{11k}}{v_{11}}-\frac{v_{22k}}{v_{22}}\right)^2 &= \frac{2}{v_n^2}\sum_k\left(\frac{v_nv_{11k}}{v_{11}}-v_{nk}-\frac{1-\phi}{2\phi}v_nv_k\right)^2\\ &\geq \frac{2}{v_n^2}\bigg(\left(\frac{v_nv_{111}}{v_{11}}-\left(1-\frac{1-\phi}{2}v_{22}\right)v_{1n}\right)^2\\ &\quad+\left(\frac{v_nv_{112}}{v_{11}}-\left(1-\frac{1-\phi}{2}v_{11}\right)v_{2n}\right)^2\\ &\quad+\left(\frac{v_nv_{11n}}{v_{11}}-v_{nn}-\frac{1-\phi}{2}v_{11}v_{22}\right)^2\bigg).\end{aligned}\tag{5.18}$$

On the other hand, (2.6), (2.7), (5.4) and the expression for v_{22k} imply

$$v_{121} = v_{112} + \frac{v_nv_{2n}}{v_{22}}$$

and

$$v_{122} = v_{22}\left(\frac{2v_{1n}}{v_n}-\frac{v_{111}}{v_{11}}-\frac{1-\phi}{\phi}\frac{v_nv_{1n}}{v_{11}}\right)+\frac{v_nv_{1n}}{v_{11}}.$$

After the terms with $k = 3, 4, \cdots, n-1$ are omitted, (5.5) yields

$$\begin{aligned}\frac{2}{v_{11}v_{22}}\sum_k v_{12k}^2 &\geq \frac{2}{v_{11}v_{22}}(v_{121}^2+v_{122}^2+v_{12n}^2)\\ &= \frac{2}{v_n^2}\bigg(\frac{v_{11}}{v_{22}}\left(\frac{v_nv_{112}}{v_{11}}+\phi v_{2n}\right)^2\\ &\quad+\frac{v_{22}}{v_{11}}\left(\frac{v_nv_{111}}{v_{11}}-(2+\phi-(1-\phi)v_{22})v_{1n}\right)^2+\phi v_{12n}^2\bigg).\end{aligned}\tag{5.19}$$

For the last term in (5.16), we restrict the sum to $k = a$ and deduce from Young's inequality that

$$\begin{aligned}&\frac{2}{v_n^2\left(v_{nn}-\frac{v_{1n}^2}{v_{11}}-\frac{v_{2n}^2}{v_{22}}\right)}\sum_{a=1}^2\sum_k\frac{1}{v_{aa}}\left(\sum_i v_iv_{aik}+\sum_i v_{ai}v_{ik}\right)^2\\ &\geq \frac{2}{v_n^2\left(v_{nn}-\frac{v_{1n}^2}{v_{11}}-\frac{v_{2n}^2}{v_{22}}\right)}\sum_{a=1}^2\frac{1}{v_{aa}}\left(\sum_i v_iv_{aia}+\sum_i v_{ai}^2\right)^2\\ &\geq \frac{4}{v_n^2}\sum_{a=1}^2\left(\sum_i v_iv_{aia}+\sum_i v_{ai}^2\right)-\frac{2(v_{11}+v_{22})}{v_n^2}\left(v_{nn}-\frac{v_{1n}^2}{v_{11}}-\frac{v_{2n}^2}{v_{22}}\right).\end{aligned}\tag{5.20}$$

Here, we have used (5.13). To calculate the terms linear in the third-order derivatives in (5.20), we express them in terms of v_{111}, v_{112} and v_{11n} as in dimension three. Taking $k = 2, n$ in (5.17) and using (5.4) and the commutation formula (2.7), we have

$$v_{222} = v_{22}\left(\frac{2v_{2n}}{v_n}-\frac{v_{112}}{v_{11}}-\frac{1-\phi}{\phi}\frac{v_nv_{2n}}{v_{22}}\right),$$

$$v_{2n2} = v_{22}\left(\frac{2v_{nn}}{v_n} - \frac{v_{11n}}{v_{11}} + \frac{1-\phi}{\phi}v_n\right) - v_n$$

and

$$v_{1n1} = v_{11n} - v_n.$$

It follows from these identities and the expressions for v_{121} and v_{122} that

$$\begin{aligned}
&\sum_{a=1}^{2}\left(\sum_i v_i v_{aia} + \sum_i v_{ai}^2\right)\\
&= v_1(v_{111}+v_{122}) + v_2(v_{121}+v_{222}) + v_n(v_{1n1}+v_{2n2})\\
&\quad + v_{11}^2 + v_{22}^2 + v_{1n}^2 + v_{2n}^2\\
&= -\frac{v_n v_{1n}}{v_{11}}\left(v_{111} + v_{22}\left(\frac{2v_{1n}}{v_n} - \frac{v_{111}}{v_{11}} - \frac{1-\phi}{\phi}\frac{v_n v_{1n}}{v_{11}}\right) + \frac{v_n v_{1n}}{v_{11}}\right)\\
&\quad - \frac{v_n v_{2n}}{v_{22}}\left(v_{112} + \frac{v_n v_{2n}}{v_{22}} + v_{22}\left(\frac{2v_{2n}}{v_n} - \frac{v_{112}}{v_{11}} - \frac{1-\phi}{\phi}\frac{v_n v_{2n}}{v_{22}}\right)\right)\\
&\quad + v_n\left(v_{11n} + v_{22}\left(\frac{2v_{nn}}{v_n} - \frac{v_{11n}}{v_{11}} + \frac{1-\phi}{\phi}v_n\right) - 2v_n\right) + v_{11}^2 + v_{22}^2 + v_{1n}^2 + v_{2n}^2\\
&= -\frac{v_n v_{1n}(v_{11}-v_{22})}{v_{11}^2}v_{111} - \frac{v_n v_{2n}(v_{11}-v_{22})}{v_{11}v_{22}}v_{112} + \frac{v_n(v_{11}-v_{22})}{v_{11}}v_{11n}\\
&\quad + 2v_{22}v_{nn} + v_{11}^2 + v_{22}^2 - 2v_n^2 + \left(1 - \frac{v_{22}}{v_{11}}(2+\phi)\right)v_{1n}^2 - \left(1 + \frac{v_{11}}{v_{22}}\phi\right)v_{2n}^2\\
&\quad + \frac{v_{22}(1-\phi)}{\phi}|\nabla v|^2,
\end{aligned} \tag{5.21}$$

in which (5.1)-(5.4) were used in the first two equalities and (5.5) in the last equality.

We now substitute (5.18)-(5.21) into (5.16) to obtain

$$\begin{aligned}
&\Delta\phi - 2\sum_k v_k\phi_k\\
&\geq \phi\left(2n+2-2\left(v_{11}+v_{22}+\frac{v_{1n}^2}{v_{11}}+\frac{v_{2n}^2}{v_{22}}\right) - 2\left(\frac{v_1^2}{v_{11}}+\frac{v_2^2}{v_{22}}\right) - 2\lambda\left(\frac{1}{v_{11}}+\frac{1}{v_{22}}\right)\right)\\
&\quad + \frac{2}{v_{11}v_{22}}\left(\frac{v_{11}+v_{22}}{v_{11}}\left(\frac{v_n v_{111}}{v_{11}}\right)^2 + \frac{v_{11}+v_{22}}{v_{22}}\left(\frac{v_n v_{112}}{v_{11}}\right)^2 + \left(\frac{v_n v_{11n}}{v_{11}}\right)^2\right.\\
&\qquad\qquad - \frac{2v_{1n}}{v_{11}}\left(2v_{11} + (1+\phi)v_{22} - \frac{1-\phi}{2}v_{22}(v_{11}+2v_{22})\right)\frac{v_n v_{111}}{v_{11}}\\
&\qquad\qquad - \frac{(1-\phi)v_{11}(2-v_{22})v_{2n}}{v_{22}}\frac{v_n v_{112}}{v_{11}}\\
&\qquad\qquad \left. + 2\left(v_{11} - v_{22} - v_{nn} - \frac{1-\phi}{2}v_{11}v_{22}\right)\frac{v_n v_{11n}}{v_{11}}\right)
\end{aligned}$$

$$
\begin{aligned}
&+\phi v_{12n}^2+\sum_{\alpha=3}^{n-1}v_{\alpha\alpha}^2+2v_{11}^2+2v_{22}^2+v_{nn}^2-4v_n^2+(3v_{22}-v_{11})v_{nn}\\
&+(1-\phi)v_{11}v_{22}(v_{nn}+2v_{22})+\frac{1}{4}(1-\phi)^2v_{11}^2v_{22}^2\\
&+\left(4-(1-\phi)v_{22}+\frac{v_{22}}{v_{11}}\big(1+\phi-(1-\phi)v_{22}\big)^2+\frac{1}{4}(1-\phi)^2v_{22}^2\right)v_{1n}^2\\
&+\left((1-\phi)v_{11}+\frac{v_{11}}{v_{22}}(1-\phi)^2+\frac{1}{4}(1-\phi)^2v_{11}^2\right)v_{2n}^2\Big).
\end{aligned}
$$

As in the three-dimensional proof, for the terms with v_{111}, v_{112} and v_{11n}, one has

$$
\begin{aligned}
&\frac{v_{11}+v_{22}}{v_{11}}\left(\frac{v_nv_{111}}{v_{11}}\right)^2+\frac{v_{11}+v_{22}}{v_{22}}\left(\frac{v_nv_{112}}{v_{11}}\right)^2+\left(\frac{v_nv_{11n}}{v_{11}}\right)^2\\
&-\frac{2v_{1n}}{v_{11}}\left(2v_{11}+(1+\phi)v_{22}-\frac{1-\phi}{2}v_{22}(v_{11}+2v_{22})\right)\frac{v_nv_{111}}{v_{11}}\\
&-\frac{(1-\phi)v_{11}(2-v_{22})v_{2n}}{v_{22}}\frac{v_nv_{112}}{v_{11}}\\
&+2\left(v_{11}-v_{22}-v_{nn}-\frac{1-\phi}{2}v_{11}v_{22}\right)\frac{v_nv_{11n}}{v_{11}}\\
=&\frac{v_{11}+v_{22}}{v_{11}}\left(\frac{v_nv_{111}}{v_{11}}-\frac{2v_{11}+(1+\phi)v_{22}-\frac{1-\phi}{2}v_{22}(v_{11}+2v_{22})}{v_{11}+v_{22}}v_{1n}\right)^2\\
&+\frac{v_{11}+v_{22}}{v_{22}}\left(\frac{v_nv_{112}}{v_{11}}-\frac{(1-\phi)v_{11}(2-v_{22})}{2(v_{11}+v_{22})}v_{2n}\right)^2\\
&+\left(\frac{v_nv_{11n}}{v_{11}}+v_{11}-v_{22}-v_{nn}-\frac{1-\phi}{2}v_{11}v_{22}\right)^2\\
&-\frac{\big(2v_{11}+(1+\phi)v_{22}-\frac{1-\phi}{2}v_{22}(v_{11}+2v_{22})\big)^2}{v_{11}(v_{11}+v_{22})}v_{1n}^2-\frac{(1-\phi)^2v_{11}^2(2-v_{22})^2}{4v_{22}(v_{11}+v_{22})}v_{2n}^2\\
&-\left(v_{11}-v_{22}-v_{nn}-\frac{1-\phi}{2}v_{11}v_{22}\right)^2\\
\geq&-\frac{\big(2v_{11}+(1+\phi)v_{22}-\frac{1-\phi}{2}v_{22}(v_{11}+2v_{22})\big)^2}{v_{11}(v_{11}+v_{22})}v_{1n}^2-\frac{(1-\phi)^2v_{11}^2(2-v_{22})^2}{4v_{22}(v_{11}+v_{22})}v_{2n}^2\\
&-\left(v_{11}-v_{22}-v_{nn}-\frac{1-\phi}{2}v_{11}v_{22}\right)^2.
\end{aligned}
$$

Dropping the nonnegative term involving v_{12n}^2 gives

$$\begin{aligned}\Delta\phi - 2\sum_k v_k\phi_k \geq\ & 2(n-3)\phi + \frac{2}{v_{11}v_{22}}\sum_{\alpha=3}^{n-1} v_{\alpha\alpha}^2 \\ & + \frac{2(v_{11}+v_{22})}{v_{11}v_{22}}\left((1-\phi)\Delta v - \sum_{\alpha=3}^{n-1} v_{\alpha\alpha}\right) \\ & + 2(1-\phi)^2\left(v_{11}+v_{22}+\frac{v_{1n}^2}{v_{11}}+\frac{v_{2n}^2}{v_{22}}\right) \\ & + \frac{(1-\phi)^2}{2(v_{11}+v_{22})}\left(\frac{(v_{22}+2)^2}{v_{11}}v_{1n}^2 + \frac{(v_{11}+2)^2}{v_{22}}v_{2n}^2\right),\end{aligned} \tag{5.22}$$

where (3.1), (5.4) and (5.5) have been used. Since $\nabla^2 v > 0$ and $0 < \phi(x_0) < 1$, the terms with $(1-\phi)\Delta v$ and $(1-\phi)^2$ are nonnegative. Thus, (5.22) yields

$$\begin{aligned}&\Delta\phi - 2\sum_k v_k\phi_k \\ &\geq 2(n-3)\phi + \frac{2}{v_{11}v_{22}}\sum_{\alpha=3}^{n-1}\left(v_{\alpha\alpha}^2 - (v_{11}+v_{22})v_{\alpha\alpha}\right) \\ &= \frac{2}{v_{11}v_{22}}\sum_{\alpha=3}^{n-1}(v_{\alpha\alpha}-v_{11})(v_{\alpha\alpha}-v_{22}) - 2(n-3)(1-\phi).\end{aligned} \tag{5.23}$$

We next estimate $(v_{\alpha\alpha}-v_{11})(v_{\alpha\alpha}-v_{22})$ for $\alpha = 3, 4, \cdots, n-1$. Recall the choice of ξ, η as the eigenvectors corresponding to the largest tangential eigenvalues of the inverse Hessian. Thus, the coordinate construction implies

$$v^{\alpha\alpha} \leq \frac{\sum_{i,j} v^{ij}v_{ia}v_{ja}}{\sum_i v_{ia}^2} = \frac{v_{aa}}{v_{aa}^2+v_{an}^2} \leq \frac{1}{v_{aa}}, \quad a = 1, 2.$$

Here, the equality is obtained from (5.2), (5.3) and $\sum_j v_{ij}v^{jk} = \delta_{ik}$. From (5.2) and (5.3),

$$v_{\alpha\alpha}v^{\alpha\alpha} + v_{\alpha n}v^{n\alpha} = 1, \quad \text{and } v_{\alpha\alpha}v^{n\alpha} + v_{\alpha n}v^{nn} = 0.$$

Thus,

$$v^{\alpha\alpha} = \frac{1}{v_{\alpha\alpha}} + \frac{v_{\alpha n}^2}{v_{\alpha\alpha}^2}v^{nn} \geq \frac{1}{v_{\alpha\alpha}}.$$

Consequently, $v_{\alpha\alpha} \geq \max\{v_{11}, v_{22}\}$ and therefore

$$(v_{\alpha\alpha}-v_{11})(v_{\alpha\alpha}-v_{22}) \geq 0.$$

Returning to (5.23), we conclude that

$$\Delta\phi - 2\sum_k v_k\phi_k \geq -2(n-3)(1-\phi). \tag{5.24}$$

Finally, we observe that

$$\lambda - 2(n-3) > \frac{(n-1)^2}{4} - 2(n-3) = \frac{(n-5)^2}{4} \geq 0$$

by Lemma 2.1. Equation (3.1) and estimate (5.24) imply

$$\begin{aligned}\Delta\big(e^{-v}(1-\phi)\big) &= e^{-v}\bigg(-\lambda(1-\phi) - \Delta\phi + 2\sum_k v_k\phi_k\bigg)\\ &\leq -e^{-v}\big(\lambda - 2(n-3)\big)(1-\phi)\\ &< 0,\end{aligned}$$

since $0 < \phi(x_0) < 1$. This contradicts (5.7) and the proof is complete. □

5.2. **Proof of Theorem 1.3 for $n \geq 4$.** We follow the continuity argument in Sections 3.2 and 4.2. Here, the minimum principle in Proposition 5.1 provides a uniform upper bound for ϕ_t along the domain deformation, which allows us to exclude degeneration of the Hessian.

Proof. Let $\{\Omega_t\}_{t\geq 0}$ and $v_t = -\log u_t$ be as in Section 2.5, with λ_t the first Dirichlet eigenvalue of Ω_t. Let ϕ_t denote the auxiliary function defined by (2.17) for v_t. By Proposition 2.5, there exists $T > 0$ such that $\nabla^2 v_t > 0$ and $\phi_t < 1$ throughout Ω_t whenever $t \geq T$. Let $t_* \in [0, T)$ be the smallest time such that both conditions hold for every $t > t_*$. As in Sections 3.2 and 4.2, it suffices to verify both conditions at $t = t_*$ and then show that $t_* = 0$.

We begin with a uniform estimate for ϕ_t. The boundary convexity estimate and the continuous boundary extension of ϕ_t are uniform for t near t_*. Since Ω_{t_*} is strictly horo-convex, there exists $\varepsilon \in (0, 1)$ independent of t such that

$$\phi_t(x) \leq 1 - \varepsilon \quad \text{on } \{x \in \Omega_t : 0 < e^{-v_t(x)} \leq \varepsilon\}$$

whenever $t > t_*$ is sufficiently close to t_*. On $\{x \in \Omega_t : e^{-v_t(x)} = \varepsilon\}$, we therefore have $e^{-v_t}(1-\phi_t) \geq \varepsilon^2$. By Proposition 5.1,

$$e^{-v_t}(1-\phi_t) \geq \varepsilon^2 \quad \text{in } \{x \in \Omega_t : e^{-v_t(x)} \geq \varepsilon\}.$$

Since the normalized eigenfunctions $u_t = e^{-v_t}$ depend smoothly on t, they are uniformly bounded above for t close to t_*. Consequently, the preceding estimates yield a constant $C \in (0, 1)$ independent of t such that

$$\phi_t \leq C \quad \text{in } \Omega_t. \tag{5.25}$$

We now estimate ϕ_t in (2.17) from below. At a fixed regular point, choose any two coordinate directions and label them $1, 2$. Rotate these directions within their plane so that $(v_t)_{12} = 0$ and rotate the remaining directions to arrange $(v_t)_\alpha = 0$, for $\alpha = 3, 4, \cdots, n-1$. Thus, the gradient belongs to the three-dimensional space spanned by the first, second and last directions. Within this space, the plane orthogonal to the gradient is tangent to the

level hypersurface. The maximum in (2.17) is therefore at least the value corresponding to this plane. Hence, by the block inverse formula and the equivalence of (2.14) and (2.16), we have

$$\begin{aligned}\phi_t &\geq \frac{\sum_{l,m\in\{1,2,n\}}(v_t)_l(v_t)_m(v_t)_{lm}}{\det\big((v_t)_{lm}\big)_{l,m\in\{1,2,n\}}}\\ &= \frac{(v_t)_n^2}{(v_t)_{11}(v_t)_{22}}\\ &\quad+\frac{(v_t)_{11}}{\det\big((v_t)_{lm}\big)_{l,m\in\{1,2,n\}}}\left((v_t)_1+\frac{(v_t)_{1n}}{(v_t)_{11}}(v_t)_n\right)^2\\ &\quad+\frac{(v_t)_{22}}{\det\big((v_t)_{lm}\big)_{l,m\in\{1,2,n\}}}\left((v_t)_2+\frac{(v_t)_{2n}}{(v_t)_{22}}(v_t)_n\right)^2\\ &\geq \frac{(v_t)_n^2}{(v_t)_{11}(v_t)_{22}},\end{aligned}$$

where $(v_t)_{12}=0$ has been used in the equality. We may therefore write the estimate in the original coordinates since these rotations preserve both $(v_t)_{11}(v_t)_{22}-(v_t)_{12}^2$ and $\sum_{k=3}^n(v_t)_k^2$. As the first two directions were arbitrary, (5.25) implies

$$\sum_{k\neq i,j}(v_t)_k^2\leq C\big((v_t)_{ii}(v_t)_{jj}-(v_t)_{ij}^2\big) \tag{5.26}$$

for all $i\neq j$ in every orthonormal frame.

For the calculations at $t=t_*$, write $v=v_{t_*}$ and $\lambda=\lambda_{t_*}$. By continuity, $\nabla^2 v\geq 0$ and (5.26) yields

$$\sum_{k\neq i,j}v_k^2\leq C(v_{ii}v_{jj}-v_{ij}^2). \tag{5.27}$$

Suppose that $\nabla^2 v$ degenerates at some $x_0\in\Omega_{t_*}$. Choose normal orthonormal coordinates at x_0 such that $\nabla^2 v(x_0)$ is diagonal. Since equation (3.1) guarantees $\Delta v>0$, its rank lies between 1 and $n-1$. All calculations below are carried out at x_0. The coordinate directions are extended as in Section 5.1.

Case 1. $\nabla^2 v$ has rank $n-1$. We may assume that $v_{ii}>0$, $i=1,2,\cdots,n-1$ and $v_{nn}=0$. Taking the indices n,i in (5.27) with $i=1,2,\cdots,n-1$ shows that $v_1=v_2=\cdots=v_{n-1}=0$. Moreover, v_{nn} attains its minimum value 0 at x_0, so $v_{nnk}=0$ and $v_{nnkk}\geq 0$. Thus, (5.10) and equation (3.1) reduce to

$$\sum_k v_{nnkk}=2\lambda+2v_n^2=2\sum_{j=1}^{n-1}v_{jj}.$$

For distinct $i, j \in \{1, 2, \cdots, n-1\}$, inequality (5.27) shows that

$$C(v_{nn}v_{ii} - v_{ni}^2) - \sum_{k\neq n,i} v_k^2$$

attains its minimum value 0 at x_0. Its second derivative in the x_j-direction therefore satisfies

$$0 \leq C(v_{ii}v_{nnjj} - 2v_{nij}^2) - 2v_{jj}^2. \tag{5.28}$$

Take $i = j + 1$ for $j = 1, 2, \cdots, n-2$ and take $i = 1$ when $j = n-1$. Then, by (5.27), (5.28) and the Cauchy-Schwarz inequality,

$$\begin{aligned}\sum_{j=1}^{n-1} v_{jj} = \frac{1}{2}\sum_{k=1}^{n} v_{nnkk} &\geq \frac{1}{2}\sum_{j=1}^{n-1} v_{nnjj} \\ &\geq \frac{1}{C}\left(\sum_{j=1}^{n-2}\frac{v_{jj}^2}{v_{j+1,j+1}} + \frac{v_{n-1,n-1}^2}{v_{11}}\right) \\ &\geq \frac{1}{C}\frac{\left(\sum_{j=1}^{n-1} v_{jj}\right)^2}{\sum_{j=1}^{n-1} v_{jj}} = \frac{1}{C}\sum_{j=1}^{n-1} v_{jj}.\end{aligned}$$

This contradicts $C < 1$ and $\sum_{j=1}^{n-1} v_{jj} > 0$.

Case 2. $\nabla^2 v$ has rank at most $n-2$. In this case, take $v_{11} > 0$ and $v_{n-1,n-1} = v_{nn} = 0$. As in Case 2 of Section 4.2, the first derivatives of $v_{n-1,n-1}$, v_{nn} and $v_{n-1,n}$ vanish at x_0. By (5.27), we also have $v_k = 0$ for $k = 1, 2, \cdots, n-2$. Consequently, the nonnegative function

$$C(v_{n-1,n-1}v_{nn} - v_{n-1,n}^2) - \sum_{k=1}^{n-2} v_k^2$$

vanishes at x_0. Its second derivative in the x_1-direction yields $0 \leq -2v_{11}^2 < 0$, again a contradiction.

We have now proved that $\nabla^2 v_{t_*} > 0$ throughout Ω_{t_*}, so ϕ_{t_*} is well defined. By continuity, (5.25) implies

$$\phi_{t_*} \leq C < 1.$$

Thus, both conditions remain valid at $t = t_*$.

As in Section 3.2, continuity and the boundary convexity estimate ensure that both conditions continue to hold for times slightly below t_* if $t_* > 0$. This contradicts the definition of t_*. Therefore, $t_* = 0$, and $\nabla^2 v > 0$ and $\phi < 1$ throughout Ω.

Finally, the argument at the end of Section 3.2 shows that u has a unique critical point, its maximum point. Every Σ_t, $0 < t < \max_\Omega u$, is smooth, closed and strictly convex. Let $\kappa_1 \leq \kappa_2 \leq \cdots \leq \kappa_{n-1}$ denote its principal curvatures. The comparison following (2.17)

yields

$$\frac{1}{\kappa_1\kappa_2} \leq \phi < 1.$$

We conclude that $\kappa_1\kappa_2 > 1$. This also implies $\kappa_2 > 1$ and hence

$$K = (\kappa_1\kappa_2)\prod_{\alpha=3}^{n-1}\kappa_\alpha > 1.$$

□

We conclude this section by weakening the strict horo-convexity assumption on Ω in Theorem 1.3 to horo-convexity.

Proof of Corollary 1.4. Let Ω_ε, u_ε and v_ε be as in the proof of Corollary 1.2. Theorem 1.3 and the estimates in Sections 4.2 and 5.2 give $\nabla^2 v_\varepsilon > 0$ and $\phi_\varepsilon < 1$ throughout Ω_ε. Local smooth convergence therefore implies $\nabla^2 v \geq 0$ in Ω and $\phi \leq 1$ wherever $\nabla^2 v > 0$.

We begin with a uniform estimate for ϕ_ε. By Proposition 2.3 and Hopf's lemma, $\nabla^2 v > 0$ and $\nabla v \neq 0$ in a connected neighborhood of $\partial\Omega$. We first show that $\phi < 1$ in this neighborhood. A geodesic ball containing Ω touches $\partial\Omega$ at a point where all boundary principal curvatures exceed 1, so Proposition 2.4 ensures that $\phi < 1$ near this point. A similar calculation retaining the gradient terms in the proofs of Propositions 4.1 and 5.1 shows

$$\Delta\phi \geq -C(1 - \phi + |\nabla\phi|), \quad 0 < \phi \leq 1,$$

where C is locally bounded. For $n \geq 4$, we use the smooth function in (2.18) as in the proof of Proposition 5.1. Since $\phi < 1$ near the boundary point chosen above, the strong maximum principle then establishes $\phi < 1$ throughout the connected boundary neighborhood.

We now use the inequality $\phi < 1$ near $\partial\Omega$ to estimate ϕ_ε on a fixed level set. Choose $t > 0$ sufficiently small that $\{0 < u \leq t\}$ lies in the preceding neighborhood. Compactness and local smooth convergence then ensure that

$$\phi_\varepsilon \leq C < 1 \quad \text{on } \Sigma_t,$$

uniformly for small $\varepsilon > 0$.

For $n = 3$, Proposition 4.1 extends this bound to $\{u > t\}$. For $n \geq 4$, we instead apply Proposition 5.1 to $e^{-v_\varepsilon}(1 - \phi_\varepsilon)$. Since $u_\varepsilon \to t > 0$ uniformly on Σ_t and $\phi_\varepsilon \leq C < 1$ there, the quantity $e^{-v_\varepsilon}(1 - \phi_\varepsilon)$ has a uniform positive lower bound on Σ_t. Proposition 5.1 gives the same lower bound in $\{u > t\}$. Together with the uniform upper bound for $u_\varepsilon = e^{-v_\varepsilon}$, this yields, after increasing C if necessary,

$$\phi_\varepsilon \leq C < 1 \quad \text{in } \{u > t\},$$

with C independent of ε.

It remains to exclude degeneration of $\nabla^2 v$ in $\{u > t\}$. By continuity, the corresponding estimates in Sections 4.2 and 5.2 now hold for v with $C < 1$, so the degeneration arguments in those sections apply. Hence, $\nabla^2 v > 0$ in $\{u > t\}$. Together with Proposition 2.3, this proves

$$\nabla^2 v > 0 \quad \text{in } \Omega.$$

□

AI use disclosure

This paper was written by the authors and was not generated by artificial intelligence. The ideas and proof strategies are the authors' own. AI tools were used primarily to explore auxiliary functions, especially in higher dimensions, and to assist with proof details and language editing. All calculations and proof details have been verified by the authors, who take full responsibility for the manuscript.

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